SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 5

Aim: Analyzing Data using Table Calculations in Tableau

a) Apply quick table calculations (running total, percent of total)
b) Create moving averages and cumulative metrics
c) Perform year-over-year growth analysis
d) Use ranking functions (Top N, Bottom N)
e) Implement window functions (WINDOW_SUM, WINDOW_AVG)
f) Customize addressing and partitioning
g) Create dynamic titles and labels using table calculations

Dataset: data_science_student_marks.csv

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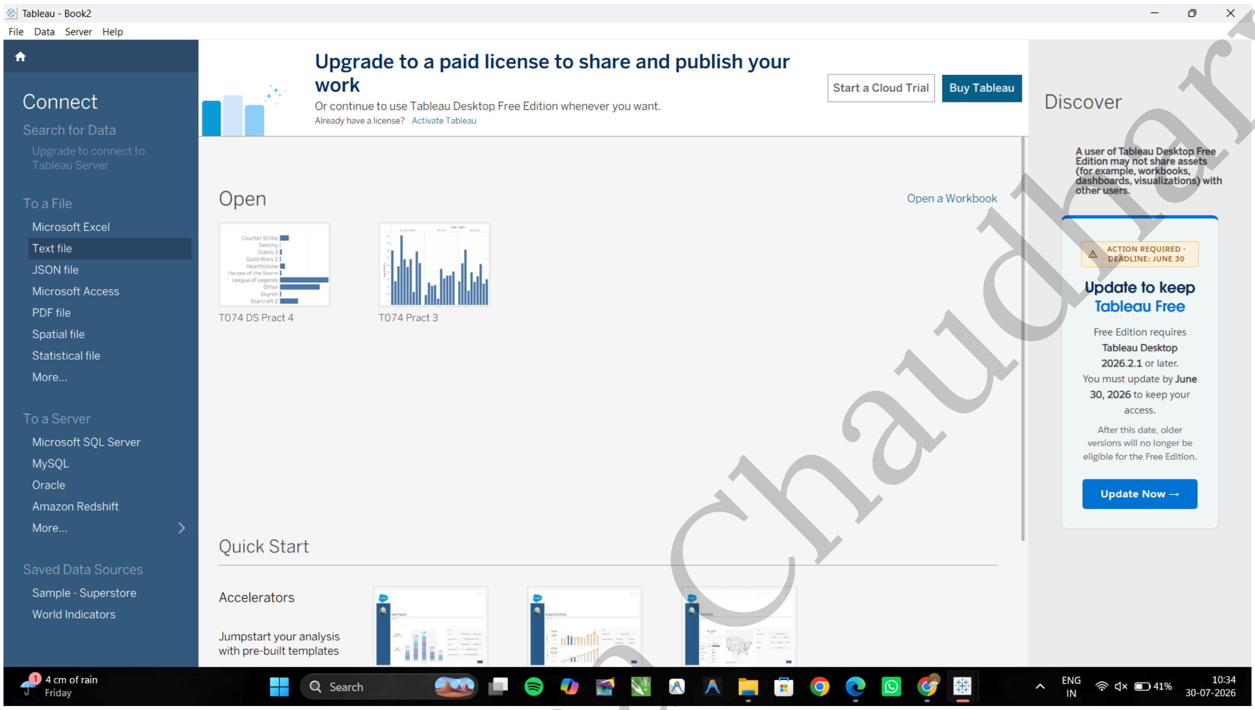
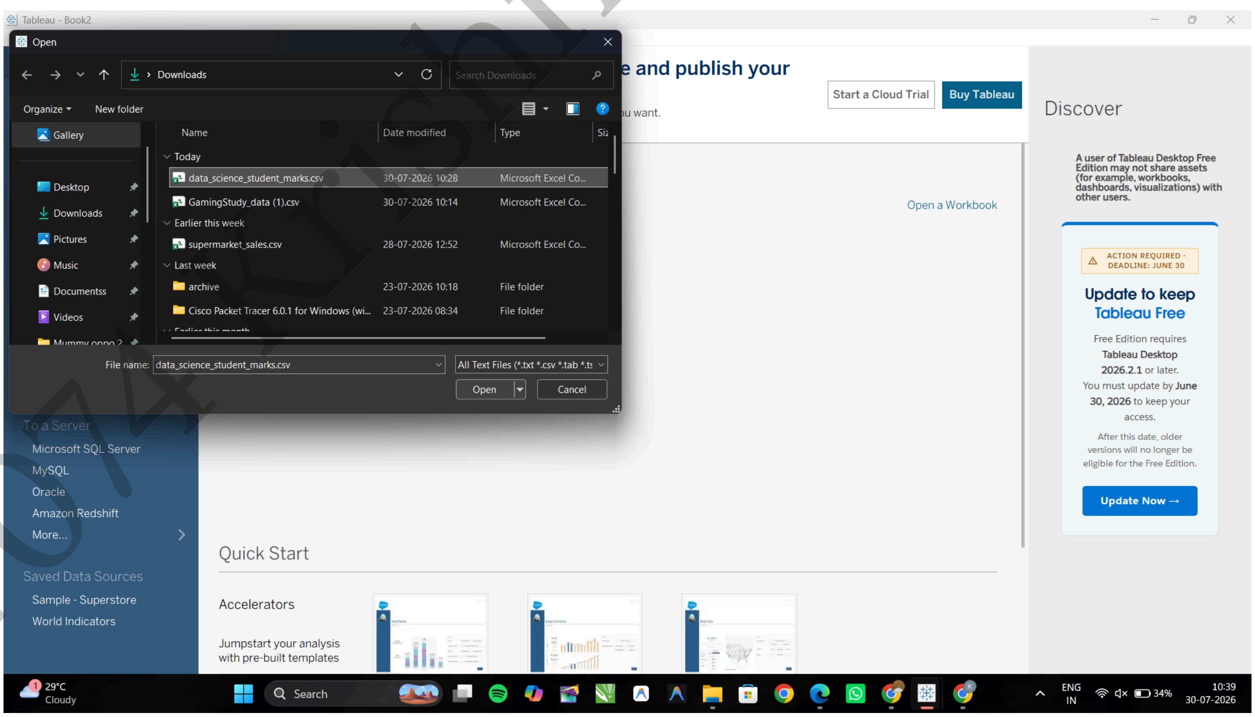
A: Apply quick table calculations (running total, percent of total)

Task 1: Create a Running Total of SQL Marks by Location

1. Launch Tableau. Under the Connect panel, click on Text File, select your Csv file, and click Open.





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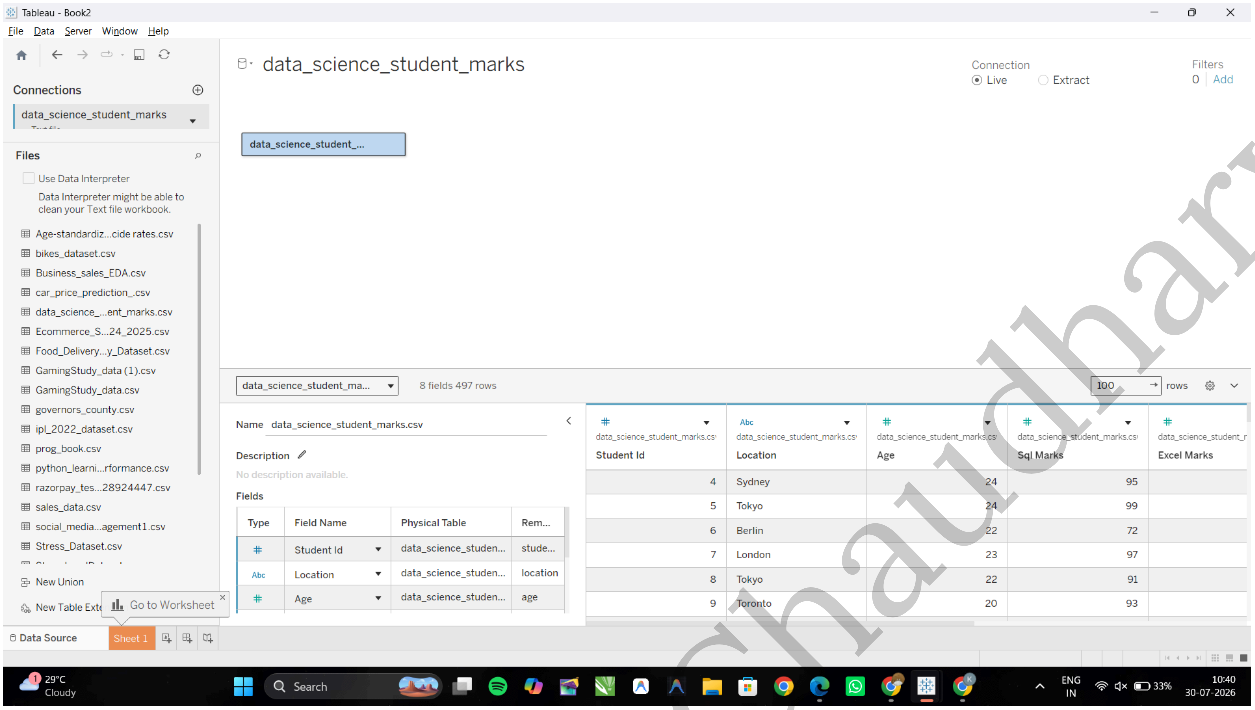
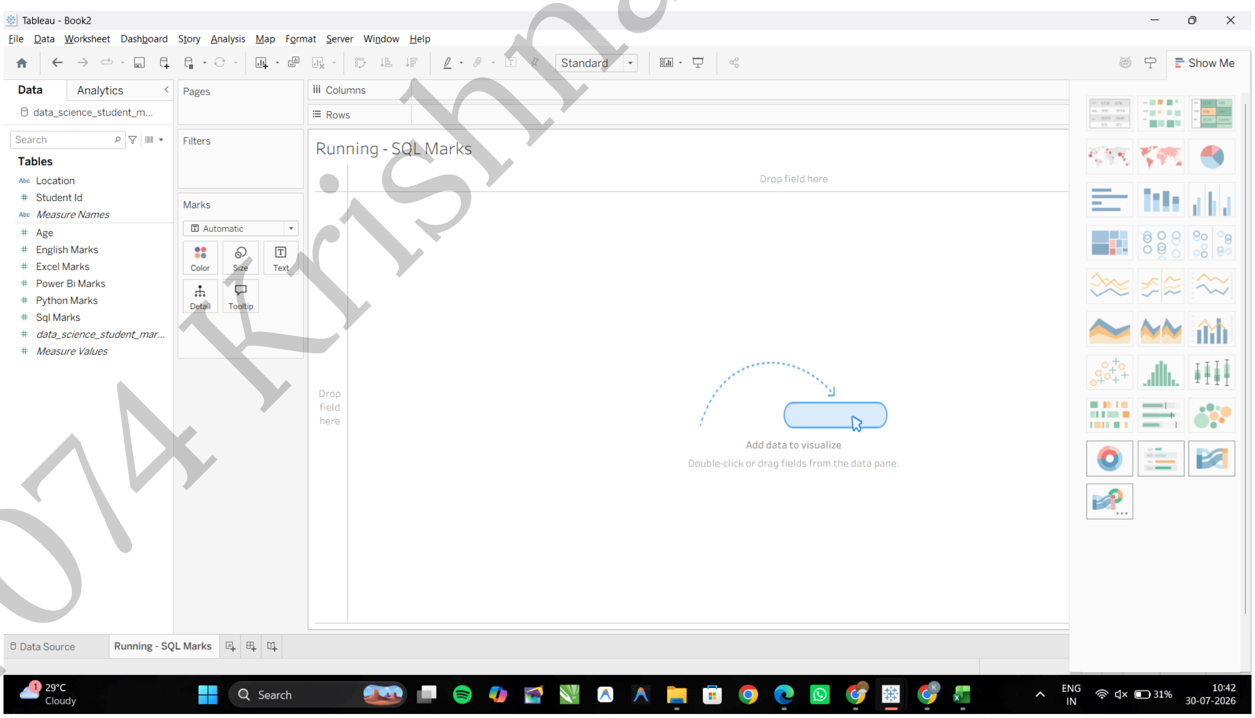


2. Open a new worksheet and rename it: Running Total - SQL Marks



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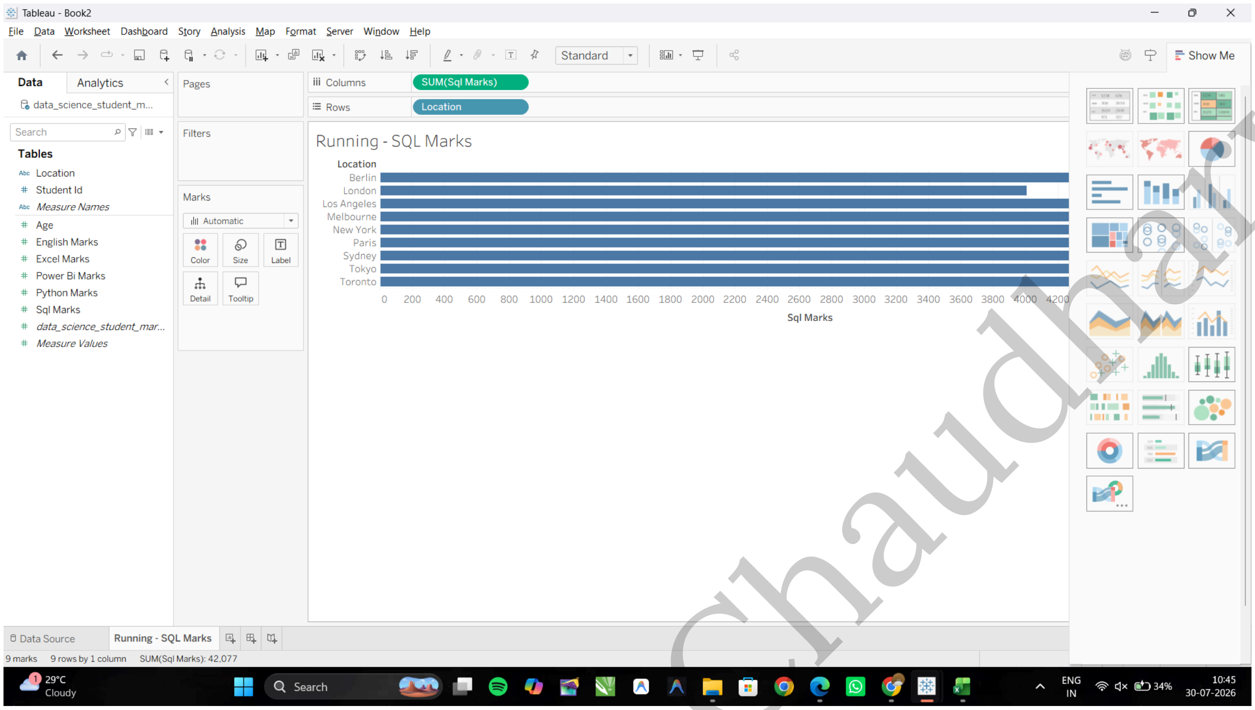
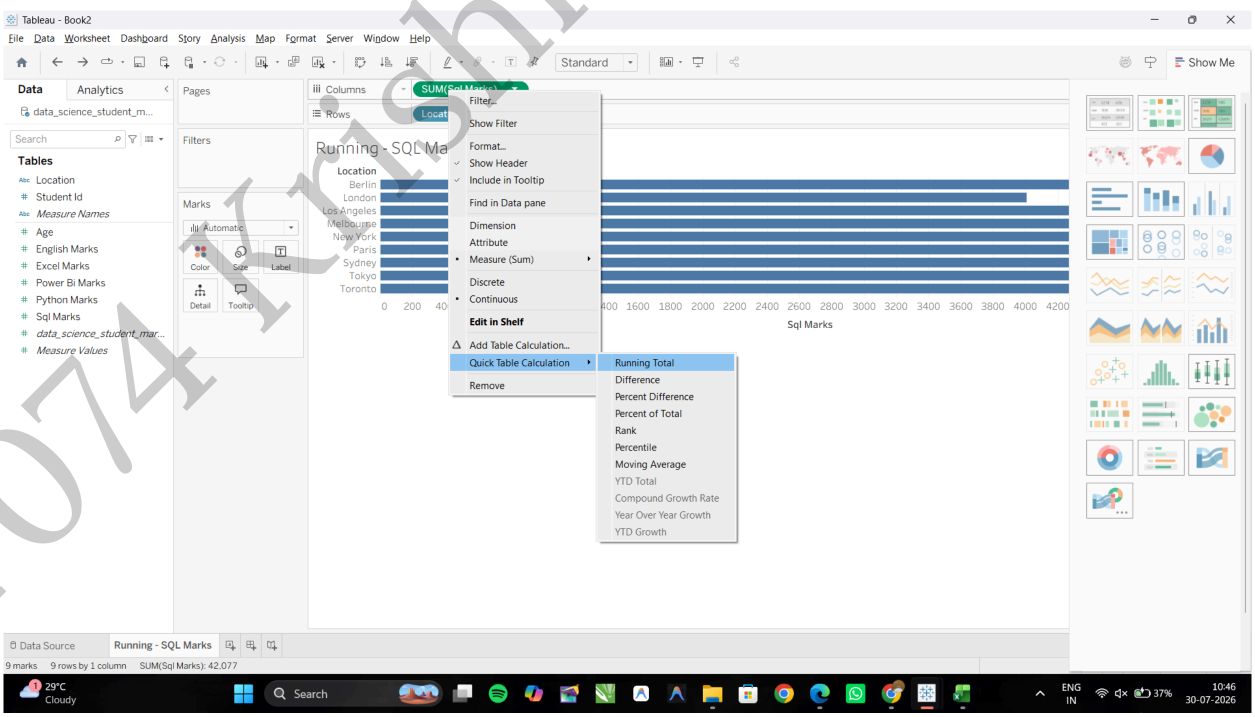
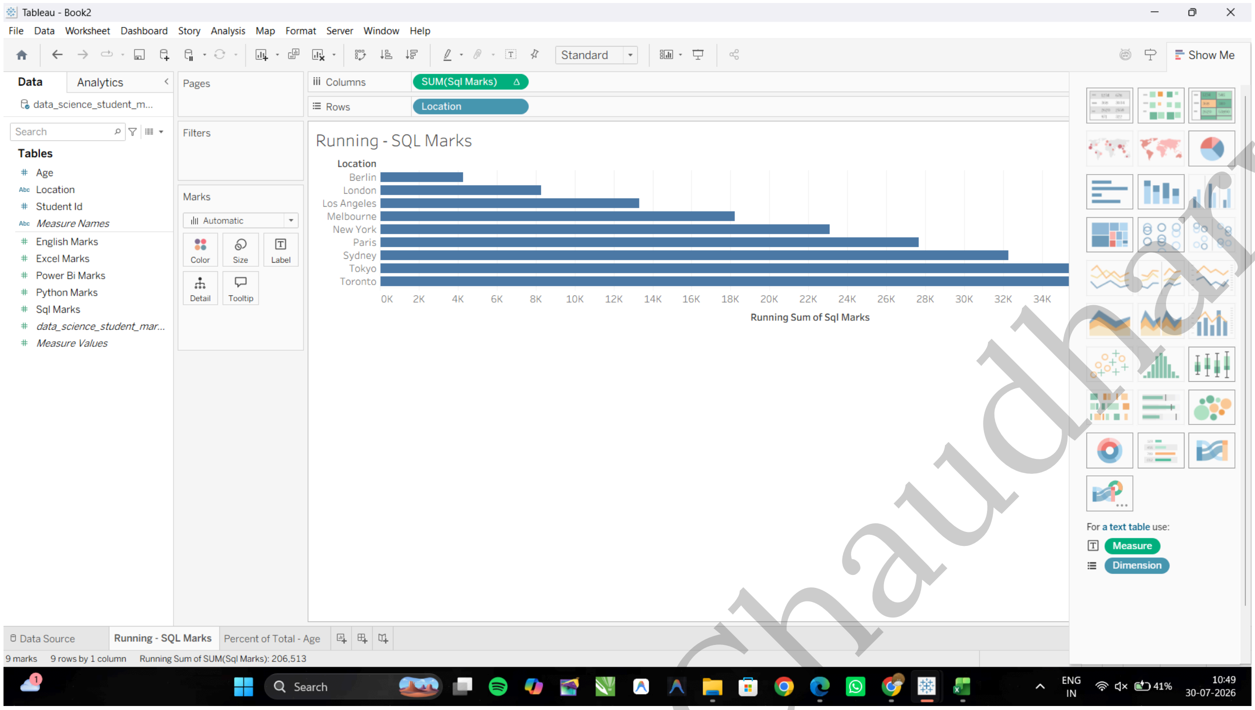
From the Data Pane, drag Location to the Rows shelf, Drag Sql Marks to the Columns shelf



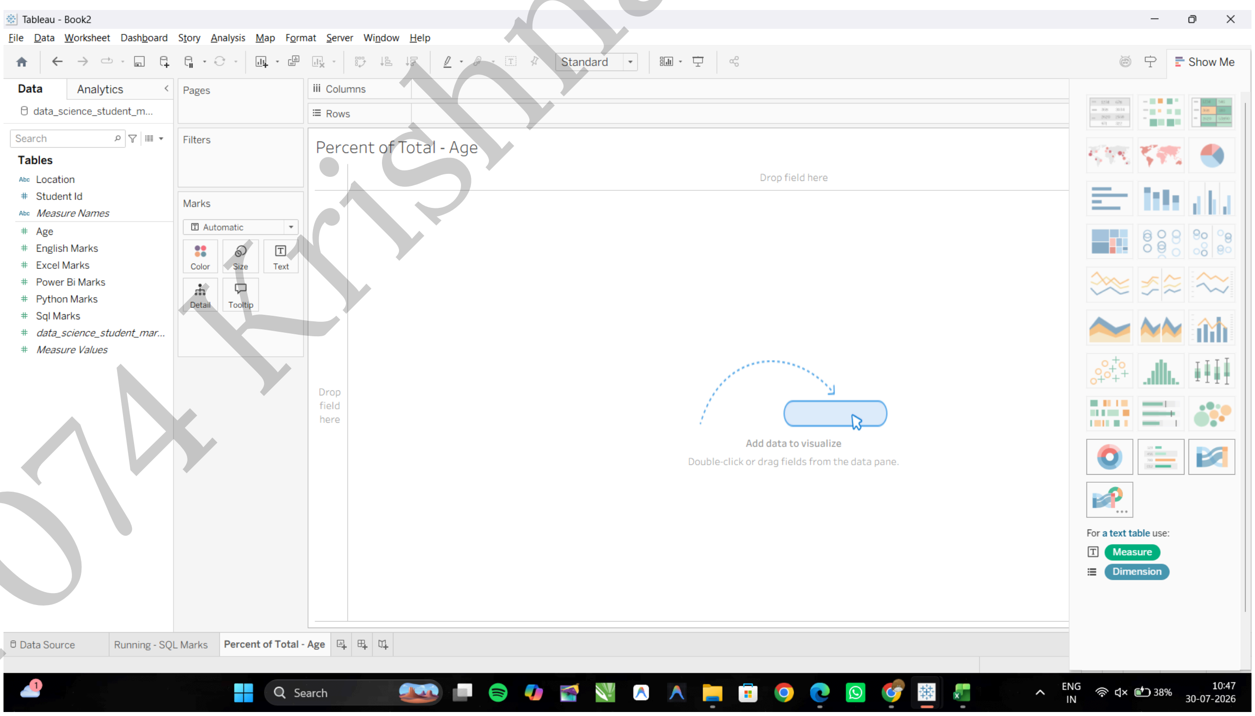
Convert to Quick Table Calculation: Right-click the SUM(Sql Marks) pill on the Columns shelf. Hover over Quick Table Calculation and select Running Total.



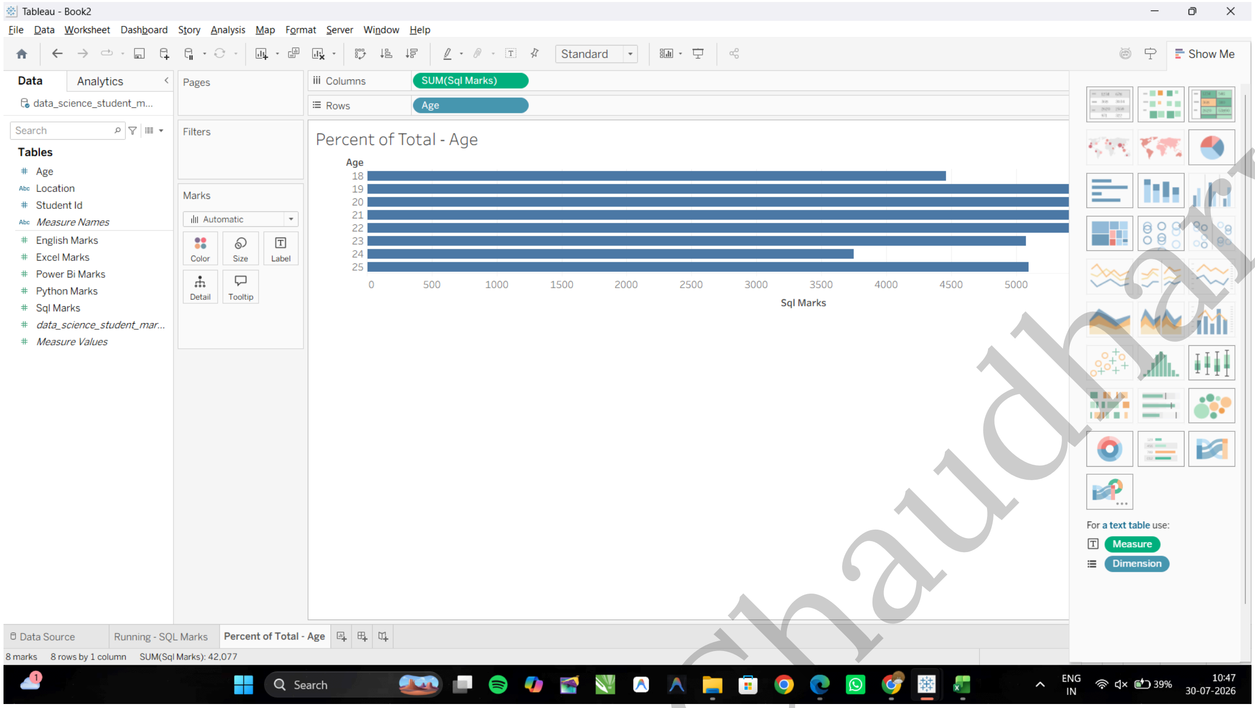


Task 2: Create a Percent of Total across Age

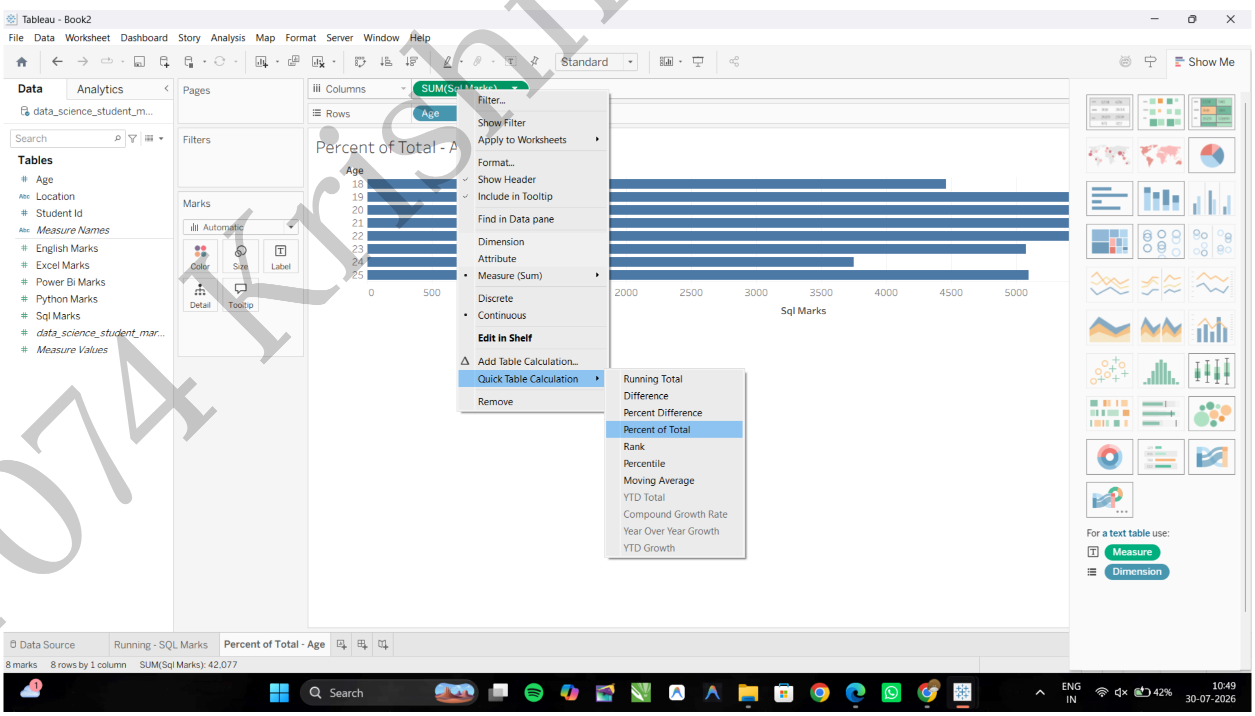
Open a new worksheet and rename it: Percent of Total - Age

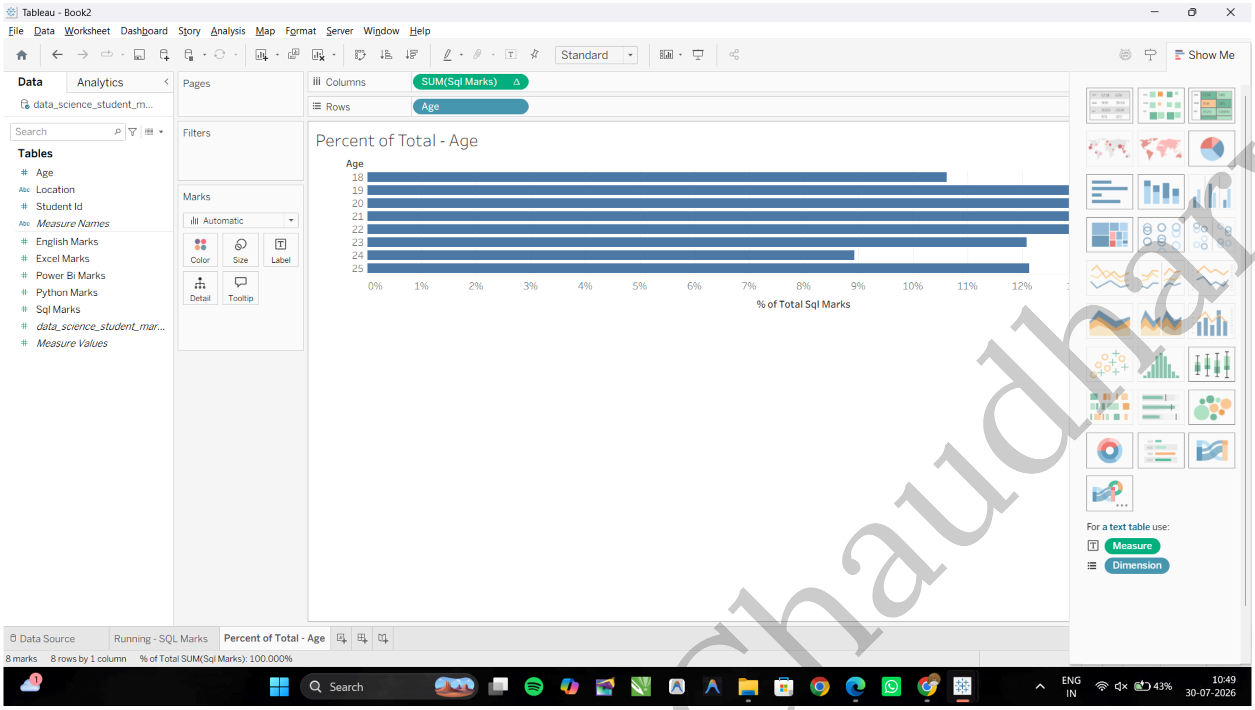


Drag Age to the Rows and Sql Marks to the Columns shelf.



Right-click the SUM(Sql Marks) pill on the Columns shelf over **Quick Table Calculation** and select **Percent of Total**.

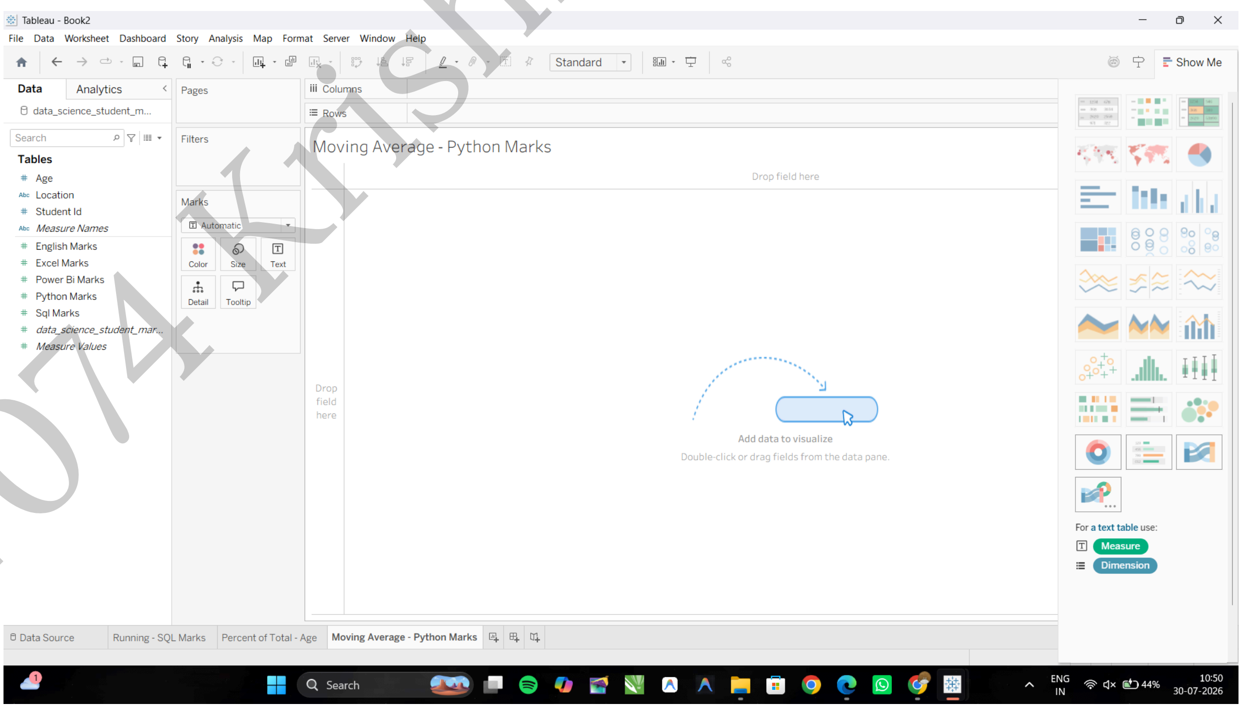




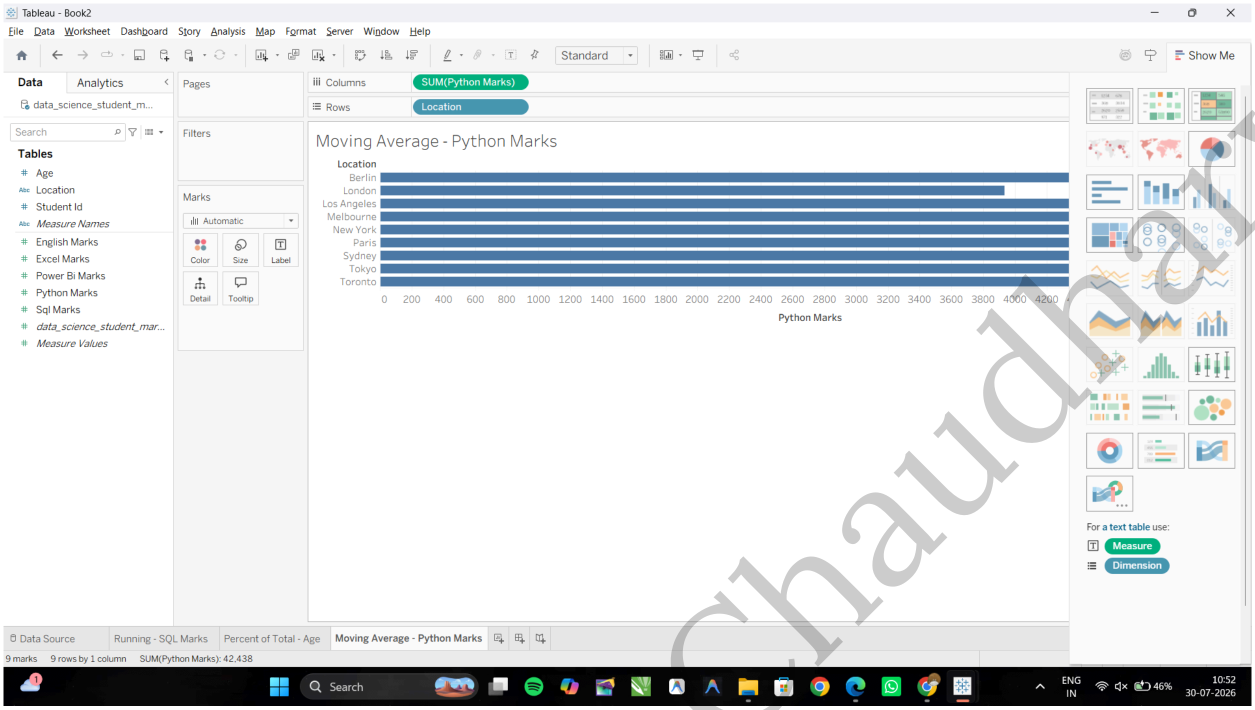
B: Create moving averages and cumulative metrics

Task 1: Create a Moving Average Calculation

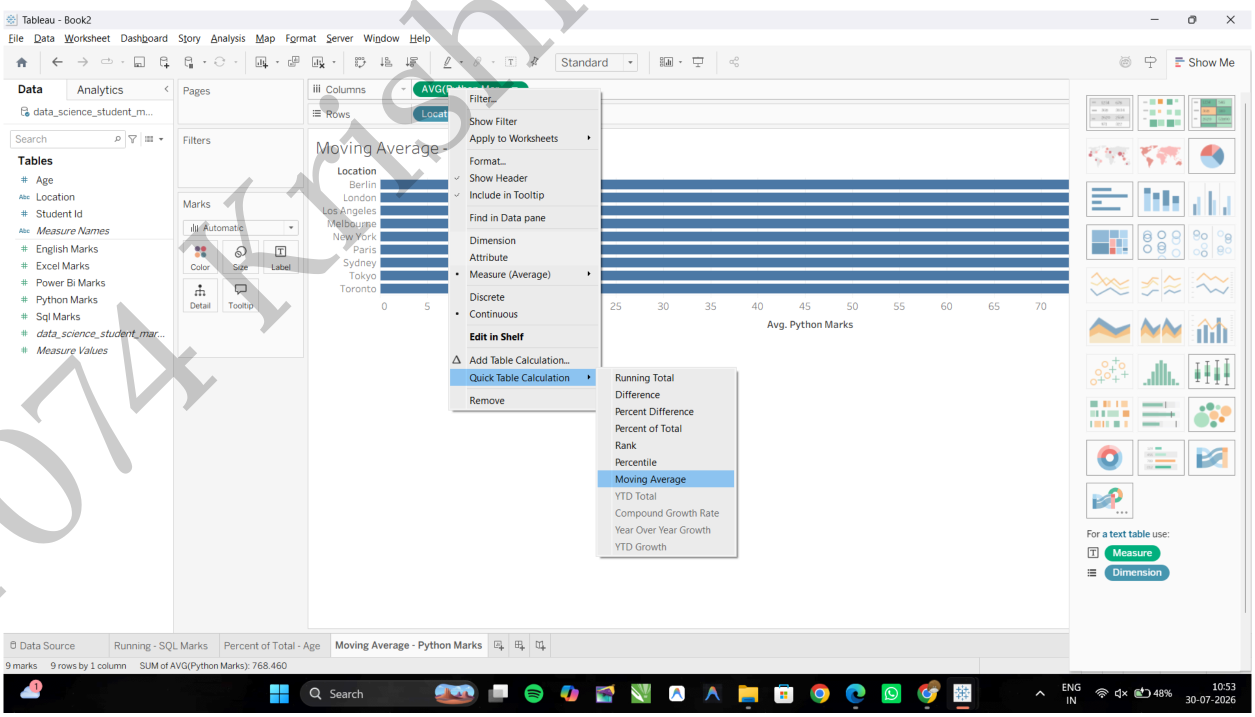
Open a new worksheet and rename it: Moving Average - Python Marks



Drag Location to the Rows and Python Marks to the Columns shelf.



Convert to a Moving Average Quick Table Calculation: Right-click the AVG(Python Marks) pill on the Columns over **Quick Table Calculation** and select **Moving Average**.



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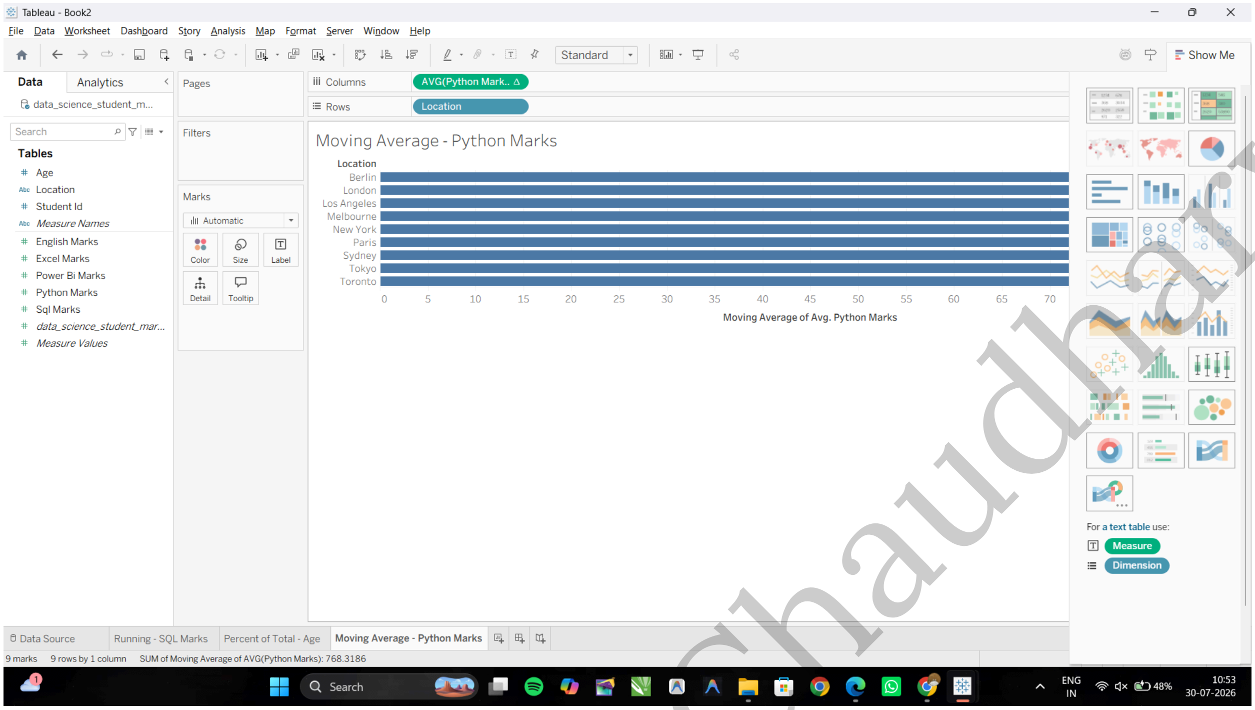
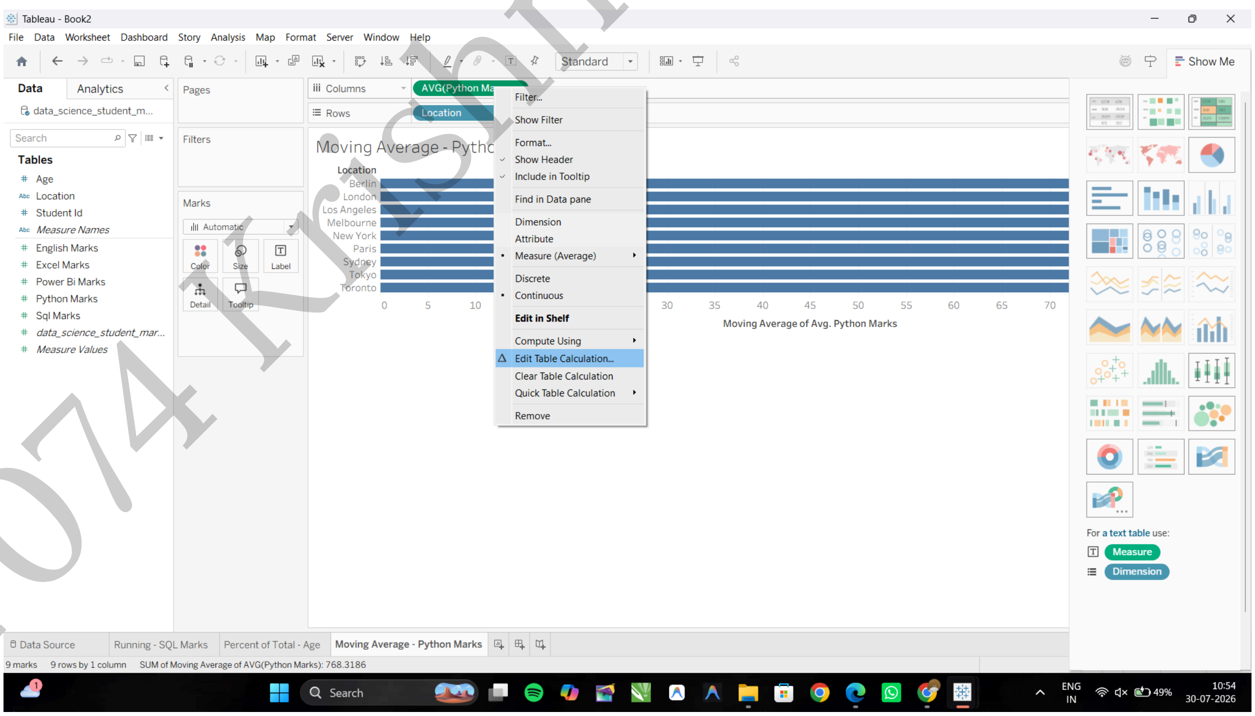


Customize the Moving Window: Right-click the modified AVG(Python Marks) pill again (it now has a Δ symbol) Select Edit Table Calculation...



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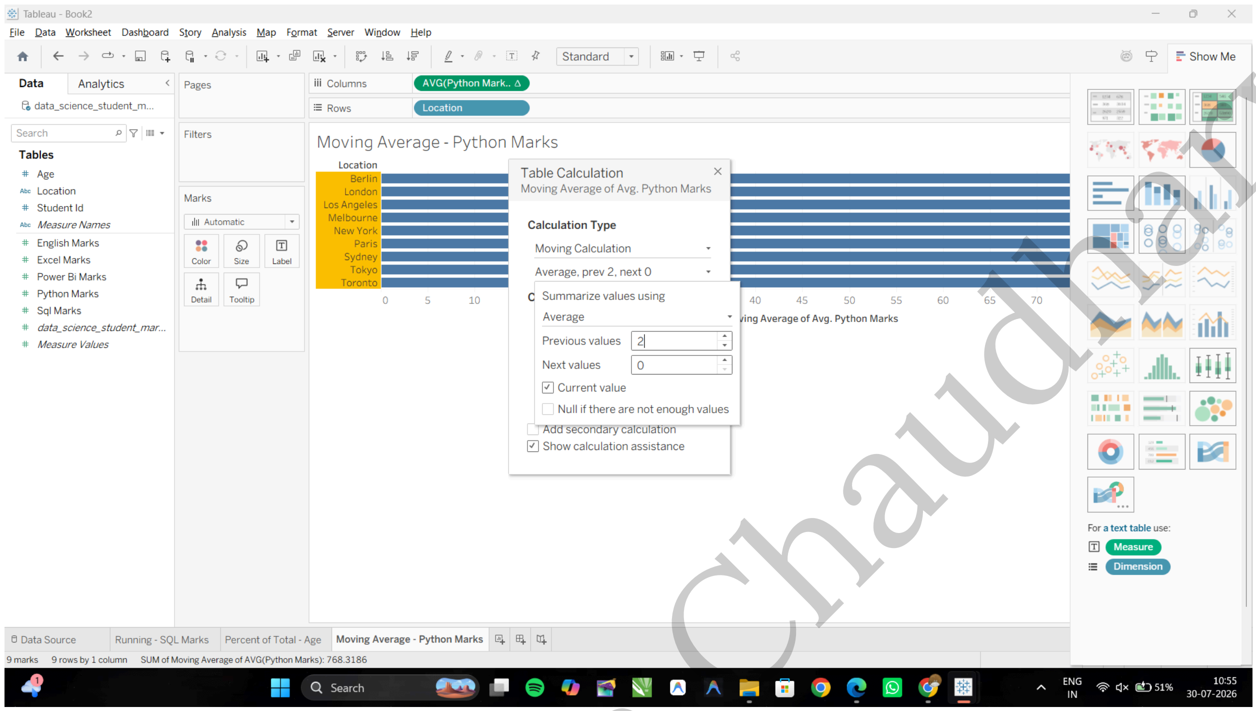
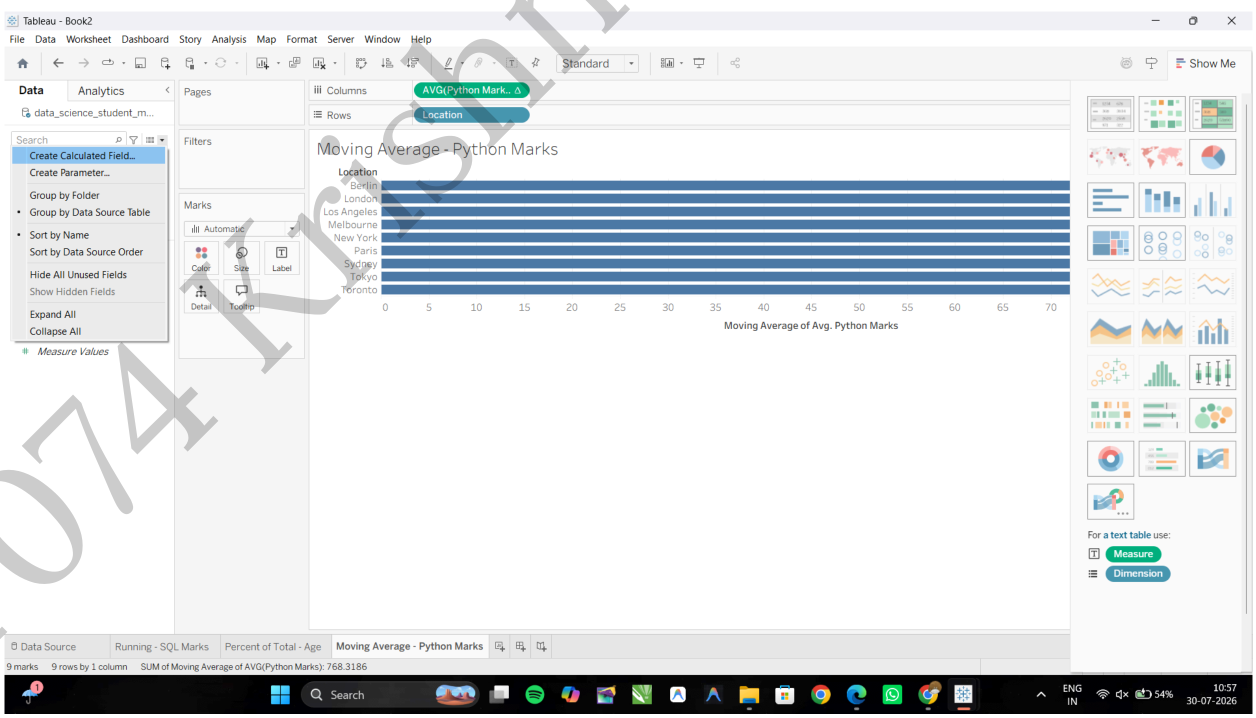
In the configuration dialog, set **Summarize values using:** Average and Set **Previous values:** 2 and **Next values:** 0 then Close the dialog box.





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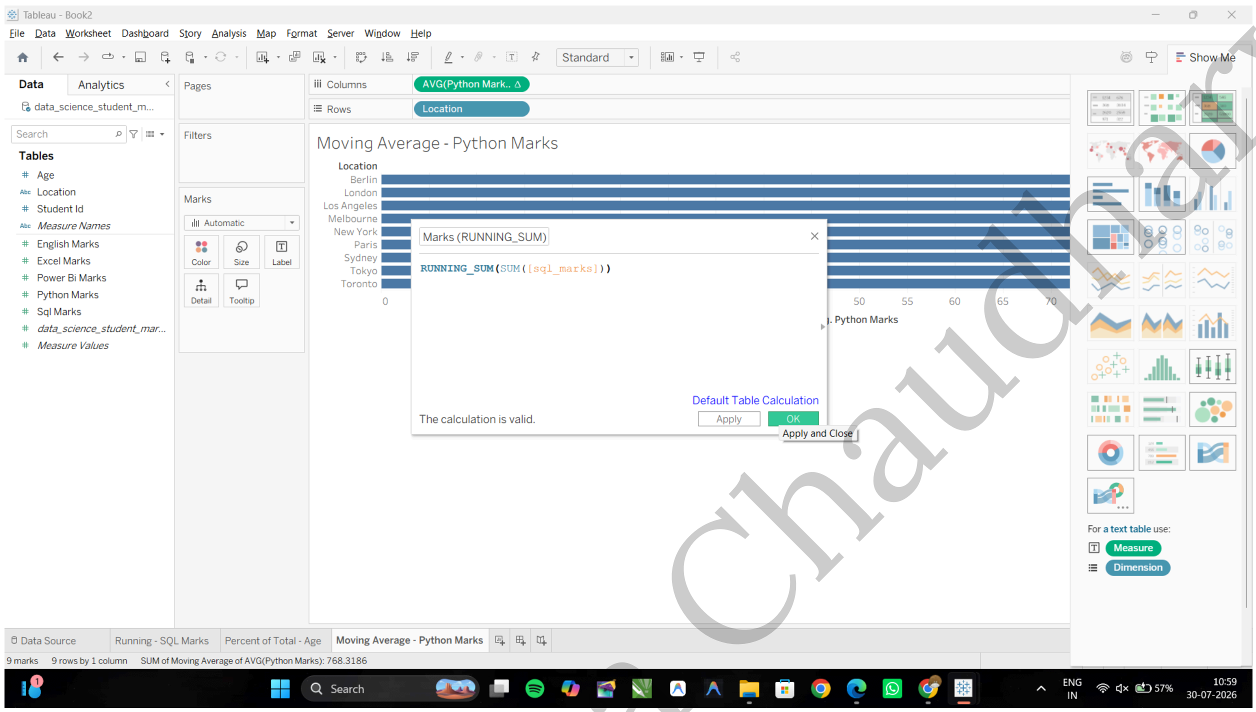
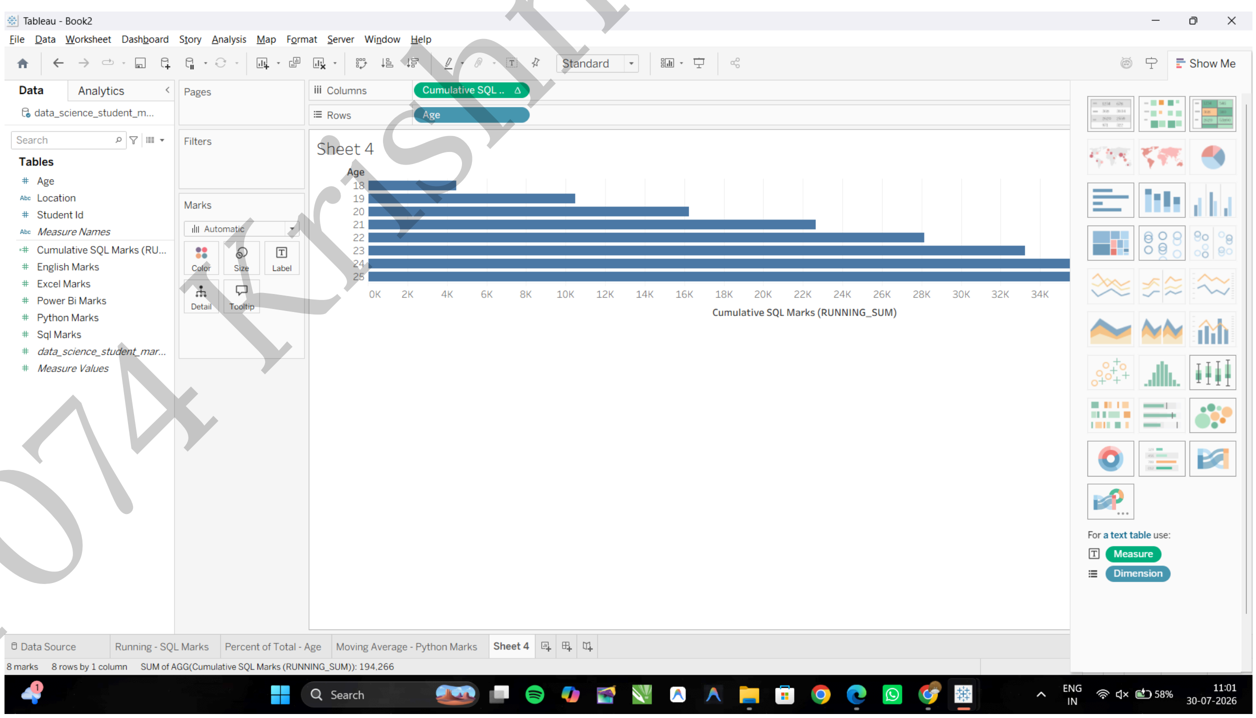
Task 2: Create a Custom Cumulative Metric via Calculated Field

Click the small drop-down arrow at the top right of the Data Pane (just next to the Search box) and select Create Calculated Field...then Name the field: Cumulative SQL Marks (RUNNING_SUM), Enter the following Table Calculation formula: RUNNING_SUM(SUM([Sql Marks])) Click OK.





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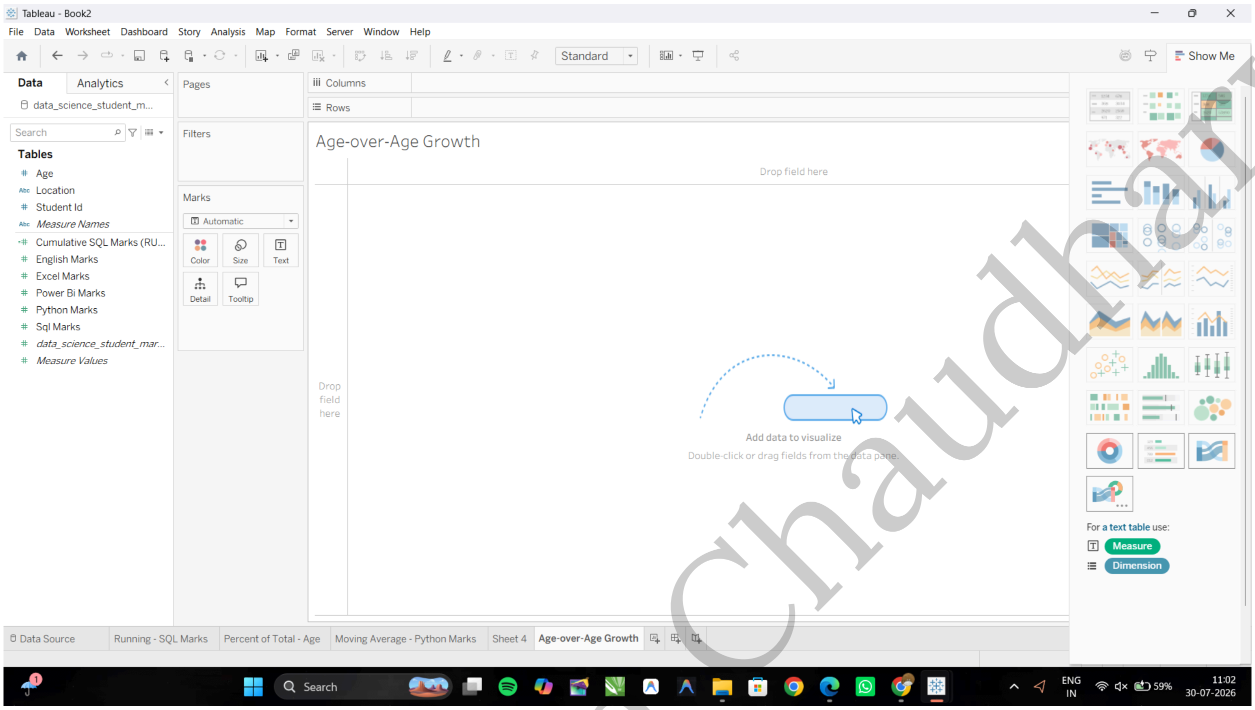
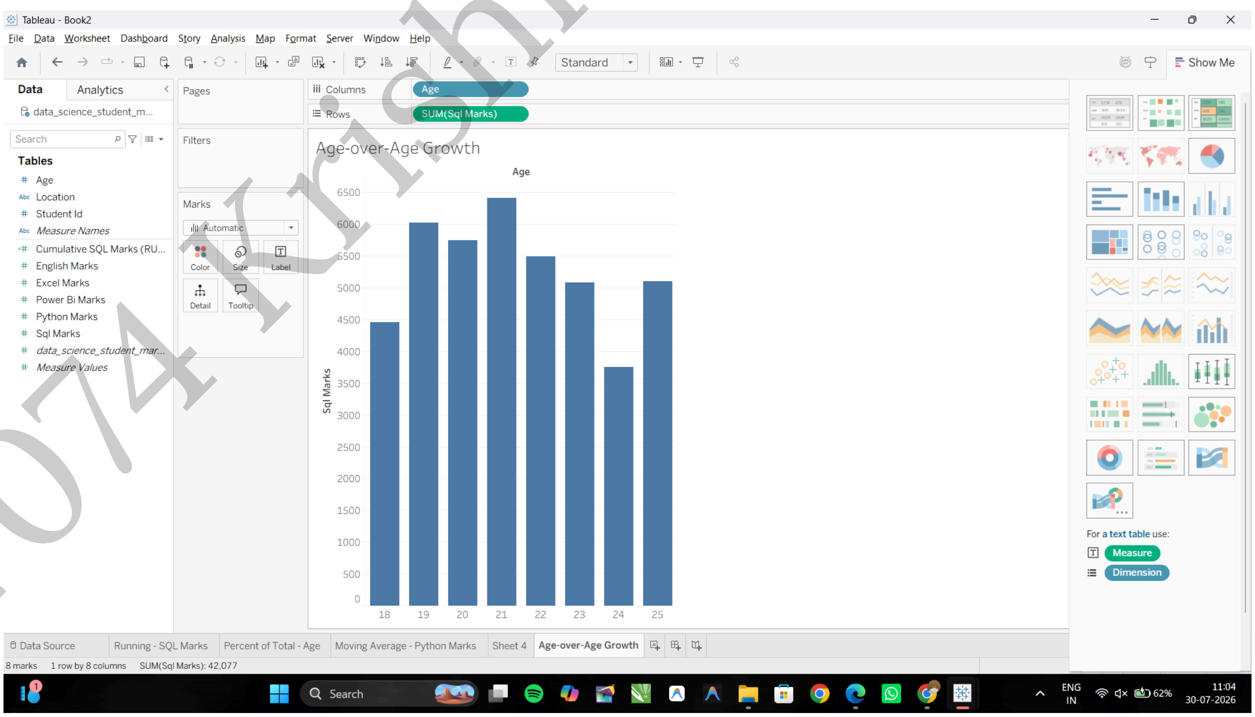
C: Perform year-over-year growth analysis

Open a new worksheet and rename it: Age-over-Age Growth



Place your time dimension (or ordinal grouped column like Age) on the Columns and Sql Marks onto the Rows shelf as SUM(Sql Marks).



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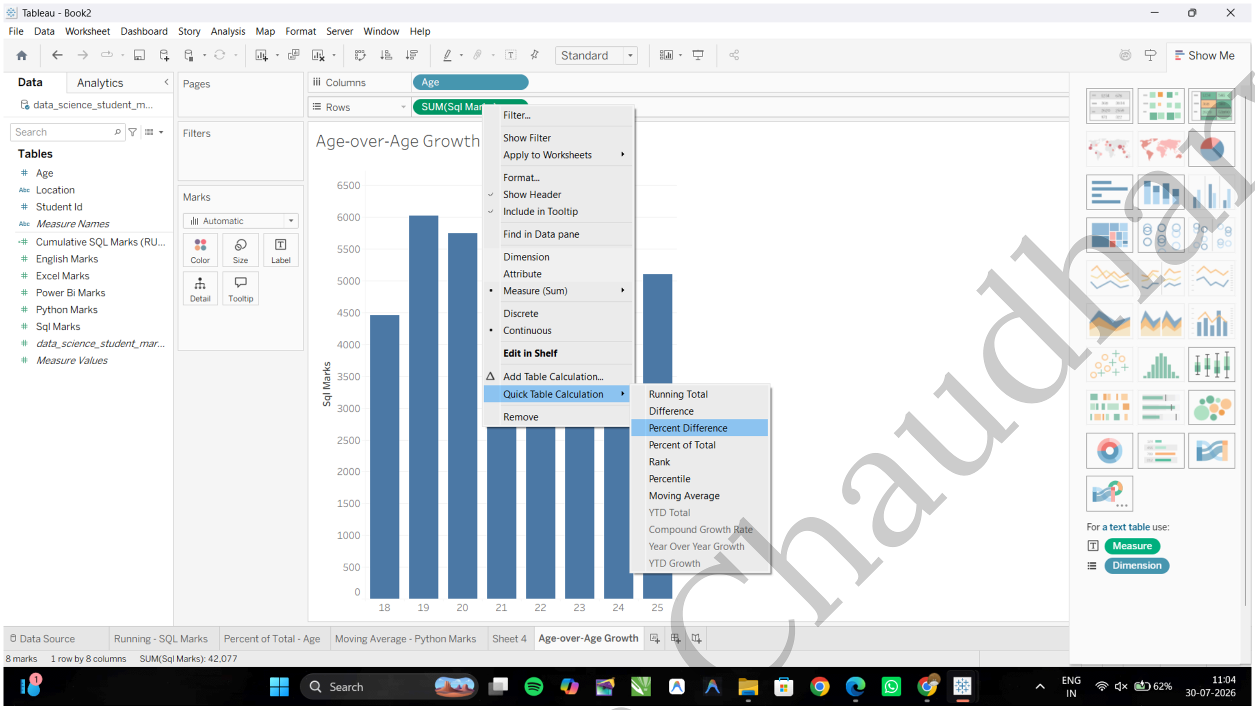
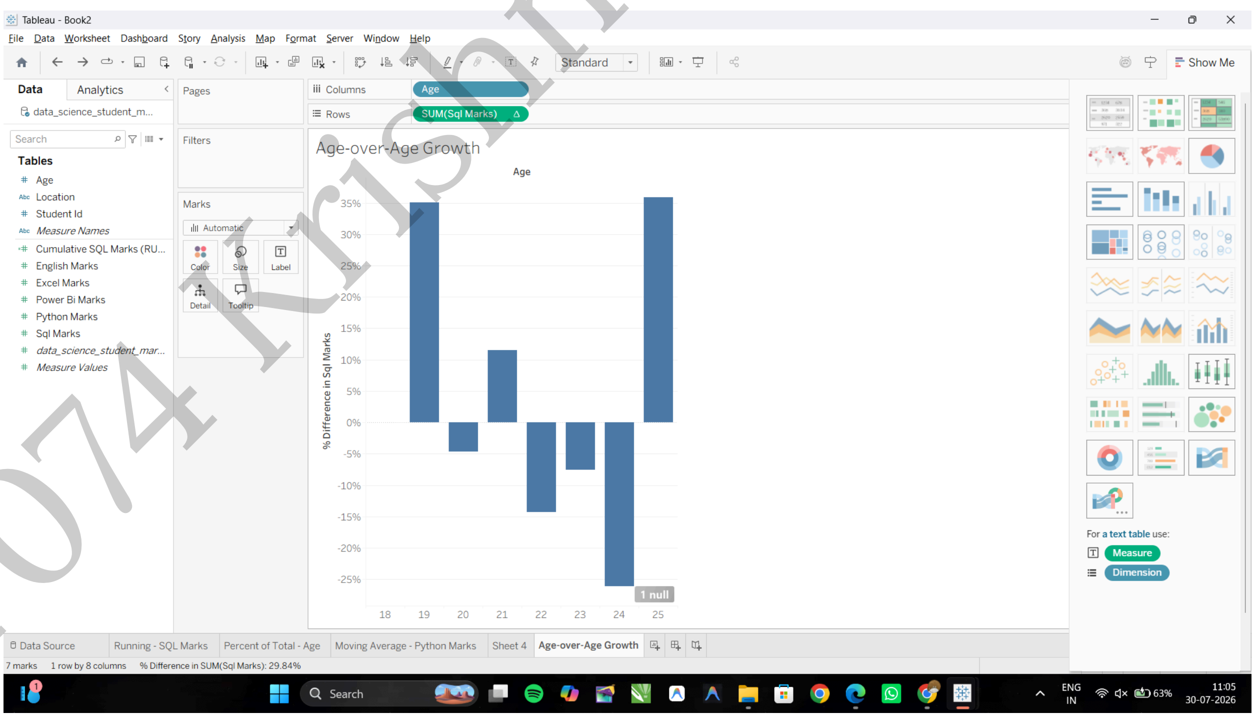
Apply the Growth Quick Table Calculation: Right-click the SUM(Sql Marks) pill on the Rows shelf and Hover over **Quick Table Calculation** → select **Percent Difference**.





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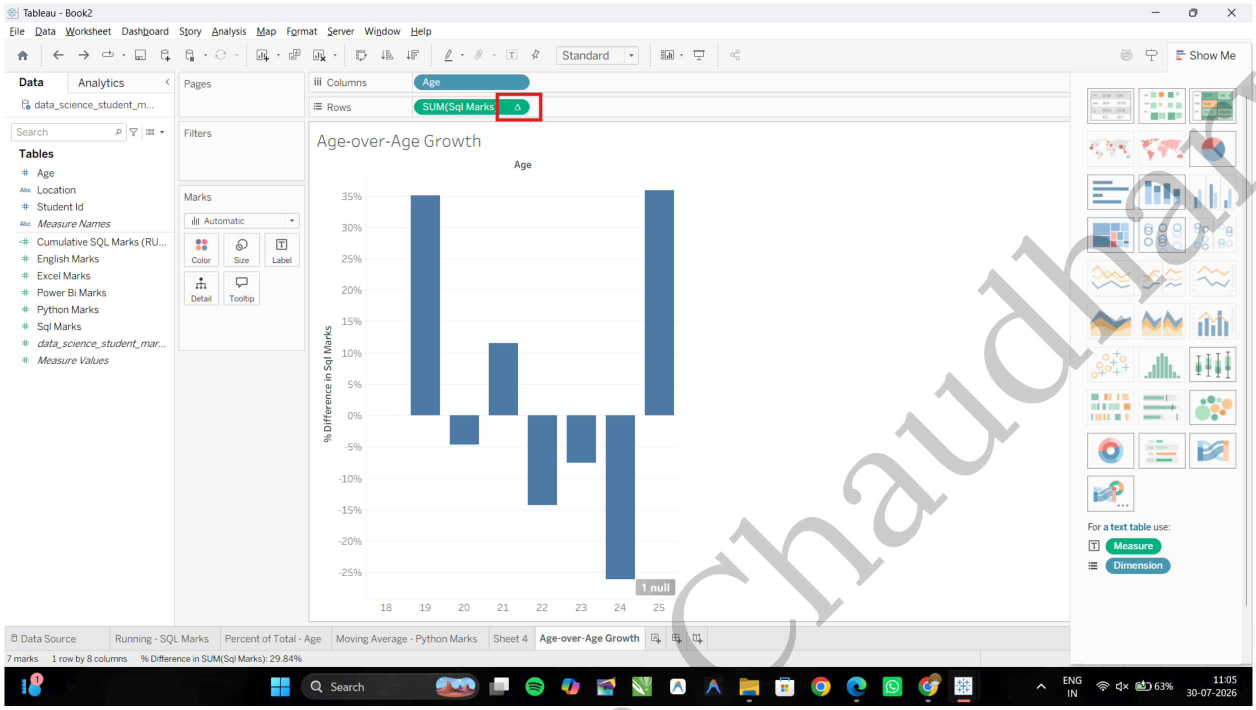
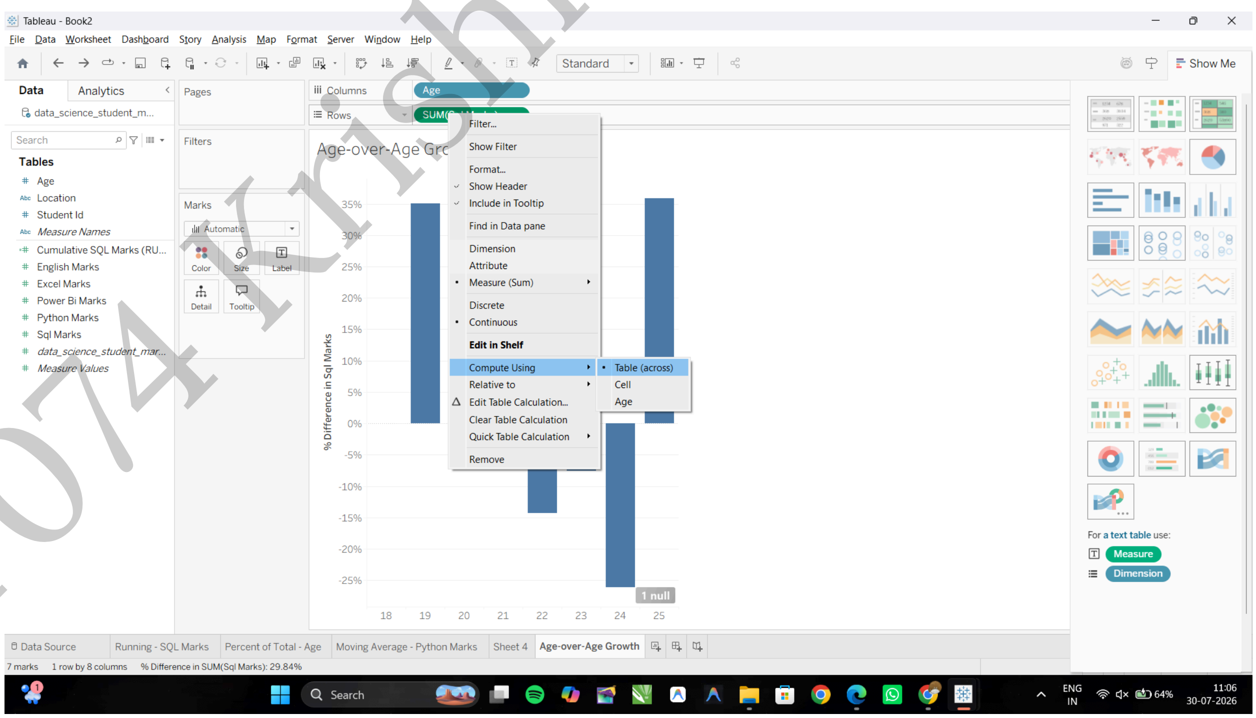
Configure the Growth Calculation Direction: Right-click the modified SUM(Sql Marks) pill (with the Δ icon)



Hover over **Compute Using** → select **Table (across)**.



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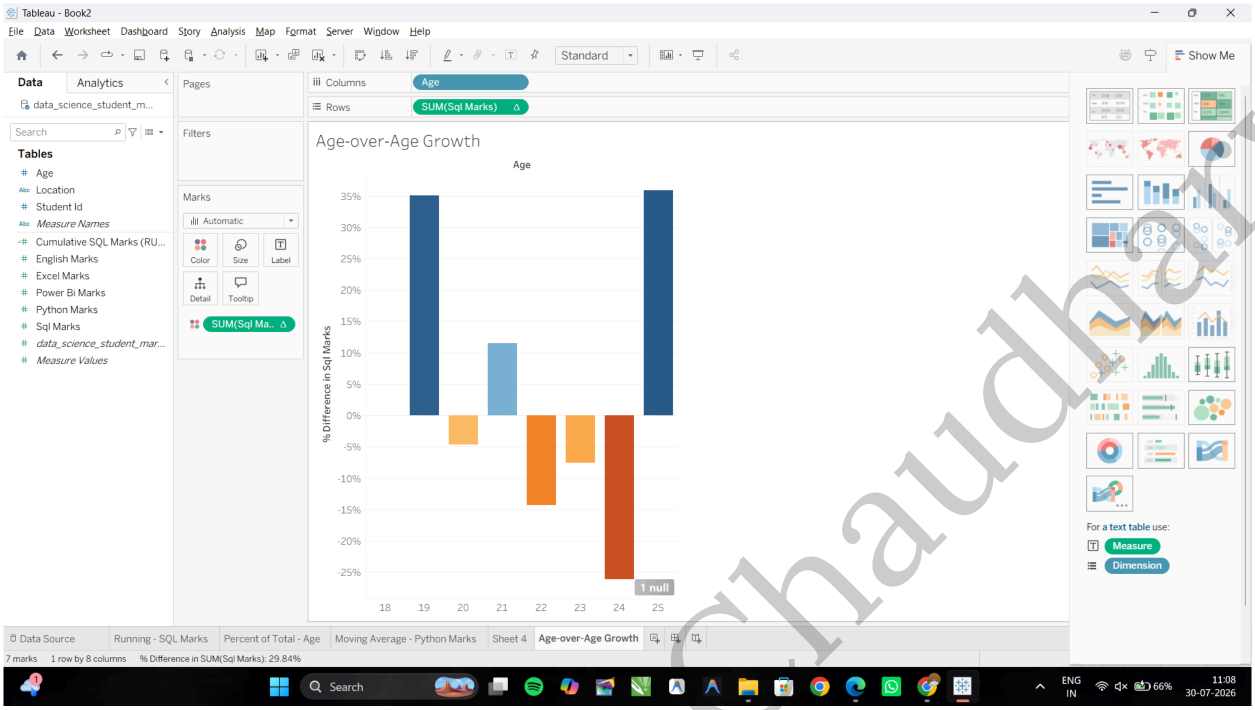
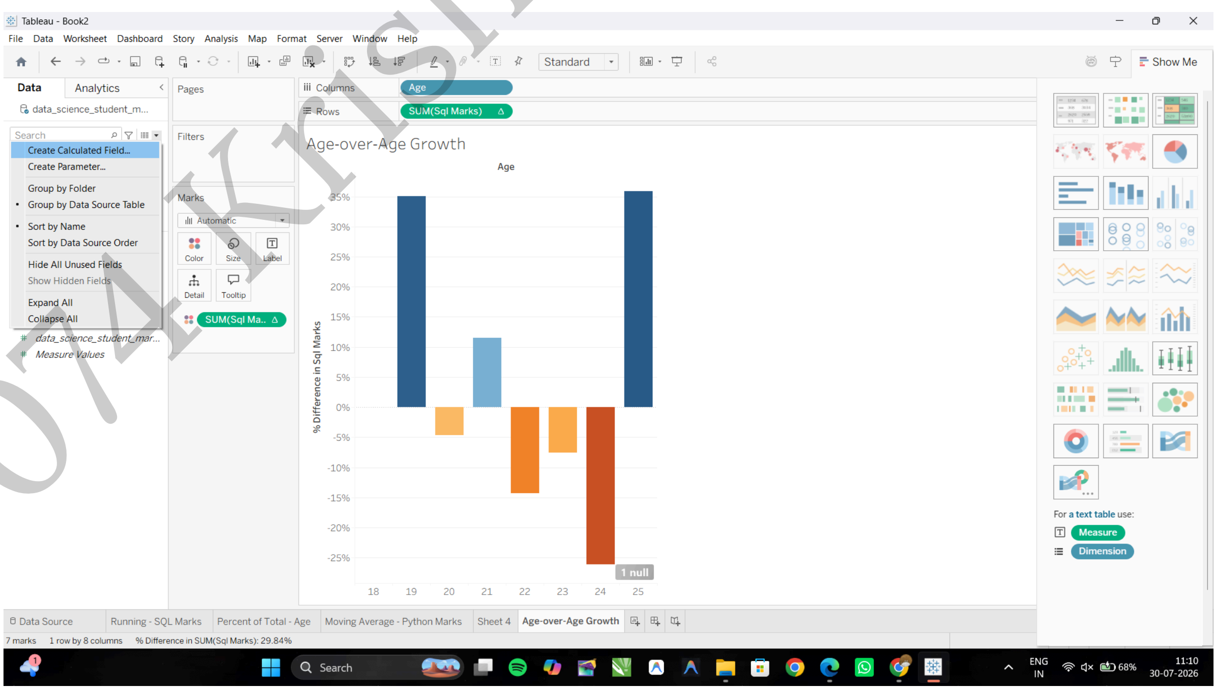
Add Color Indicators: Hold Ctrl (or Cmd on Mac), drag SUM(Sql Marks) from Rows to Color on the Marks card. Positive values show one color, while negative values appear Red/Orange.



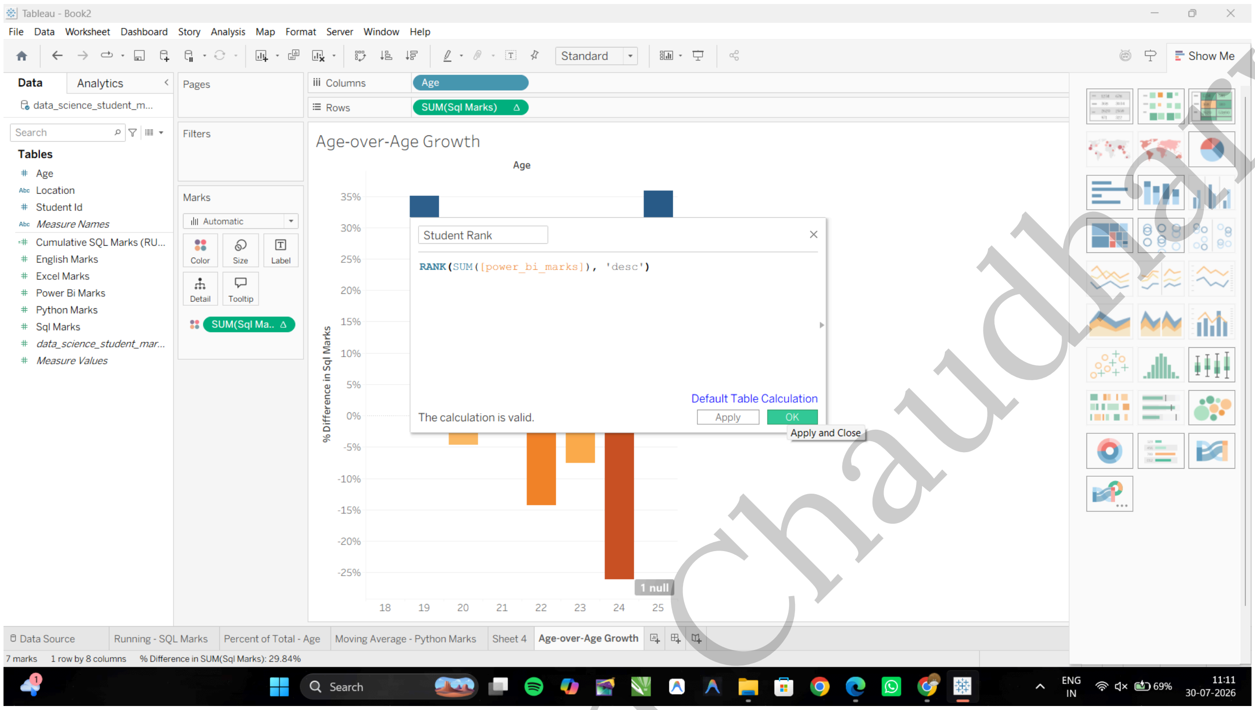
D: Use ranking functions (Top N, Bottom N)

Task 1: Create a Rank Calculation

Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...

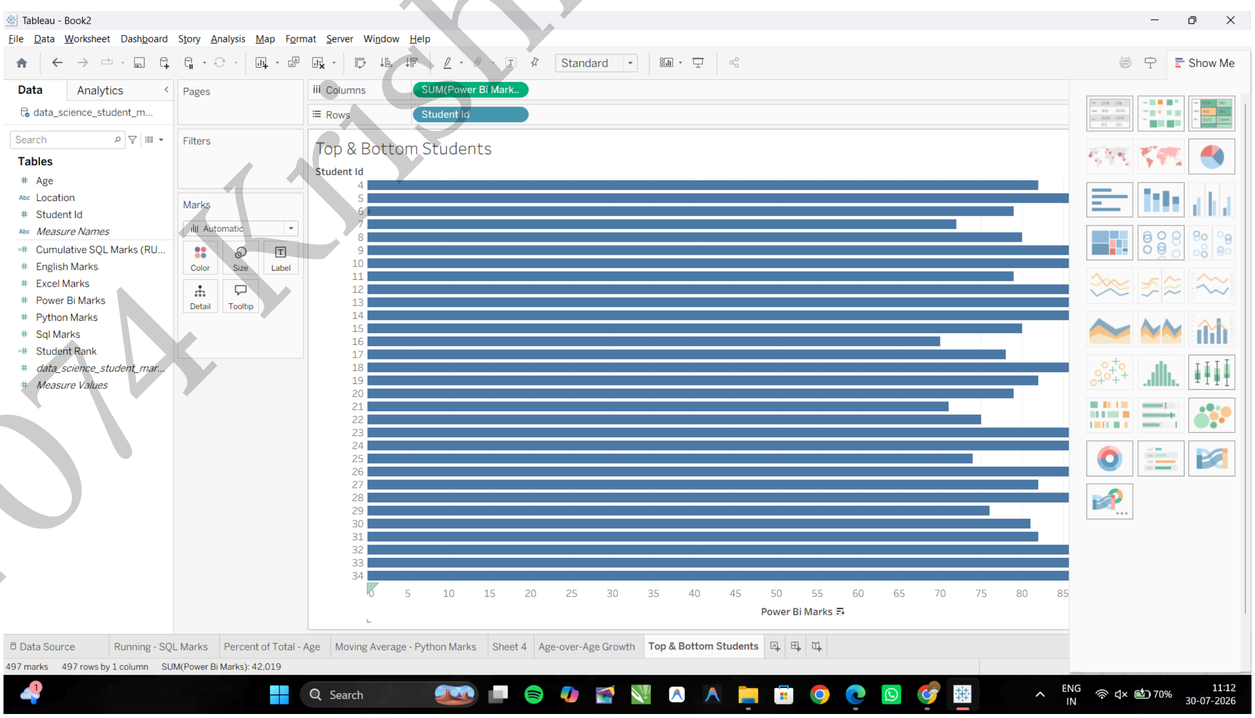


Name the field: Student Rank and Input the following ranking calculation: RANK(SUM([Sql Marks]), 'desc') then Click **OK**.



Task 2: Build the Top N/ Bottom N Ranked View

Open a new worksheet and rename it: Top & Bottom Students and Drag Student Id to the Rows Sql Marks to the Columns shelf.



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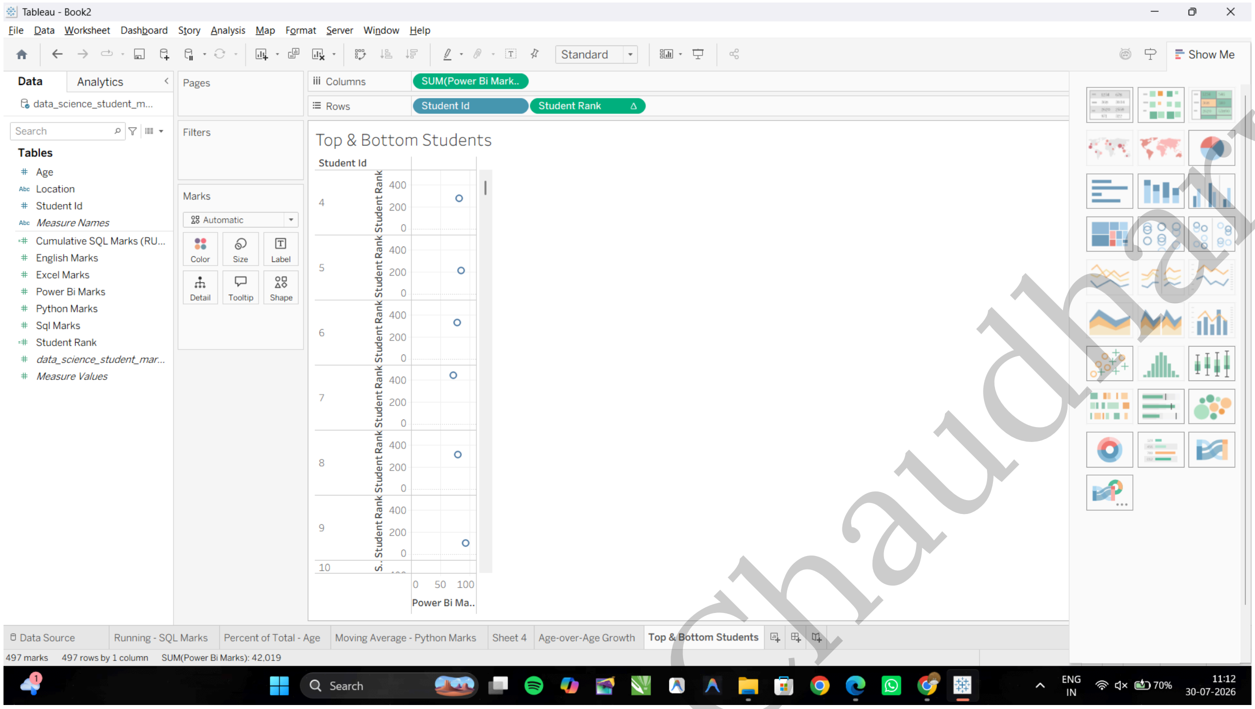
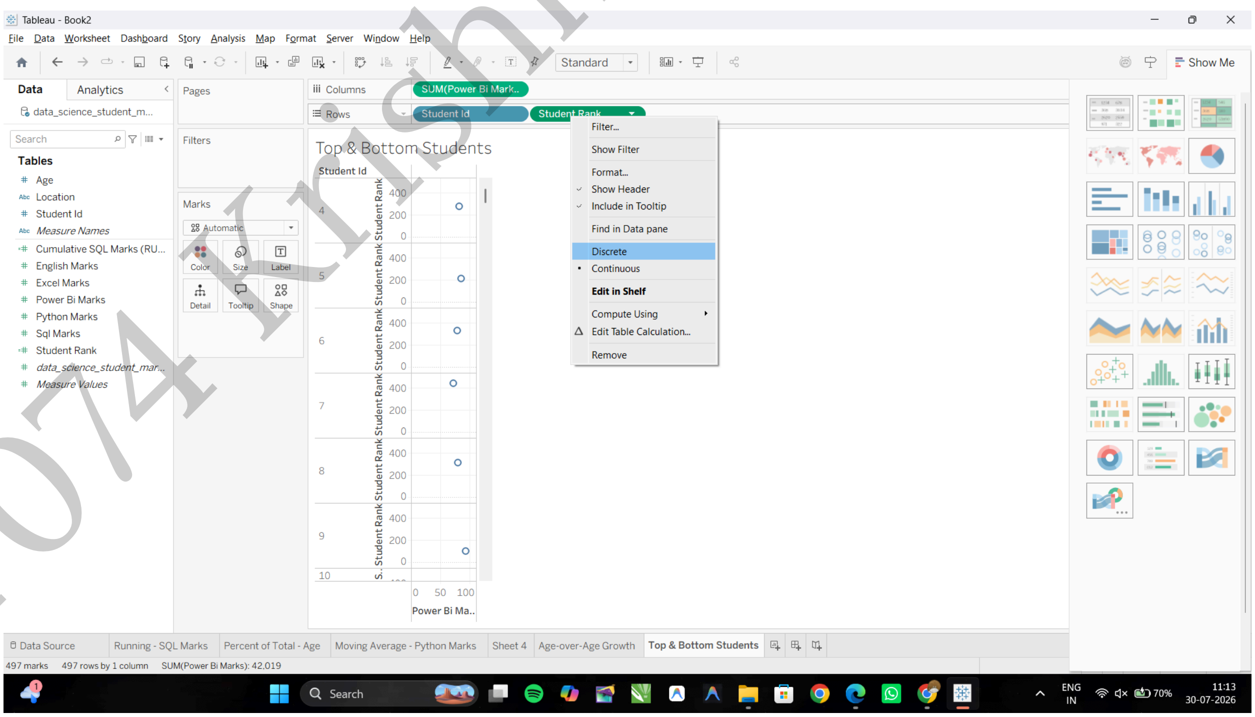
Drag your newly created Student Rank calculated field onto the Rows shelf, placing it to the left of Student Id



Convert Student Rank to a Discrete field: Right-click the Student Rank pill on the Rows shelf and select **Discrete**. (The pill will turn blue, ordering the students automatically from #1 onward).



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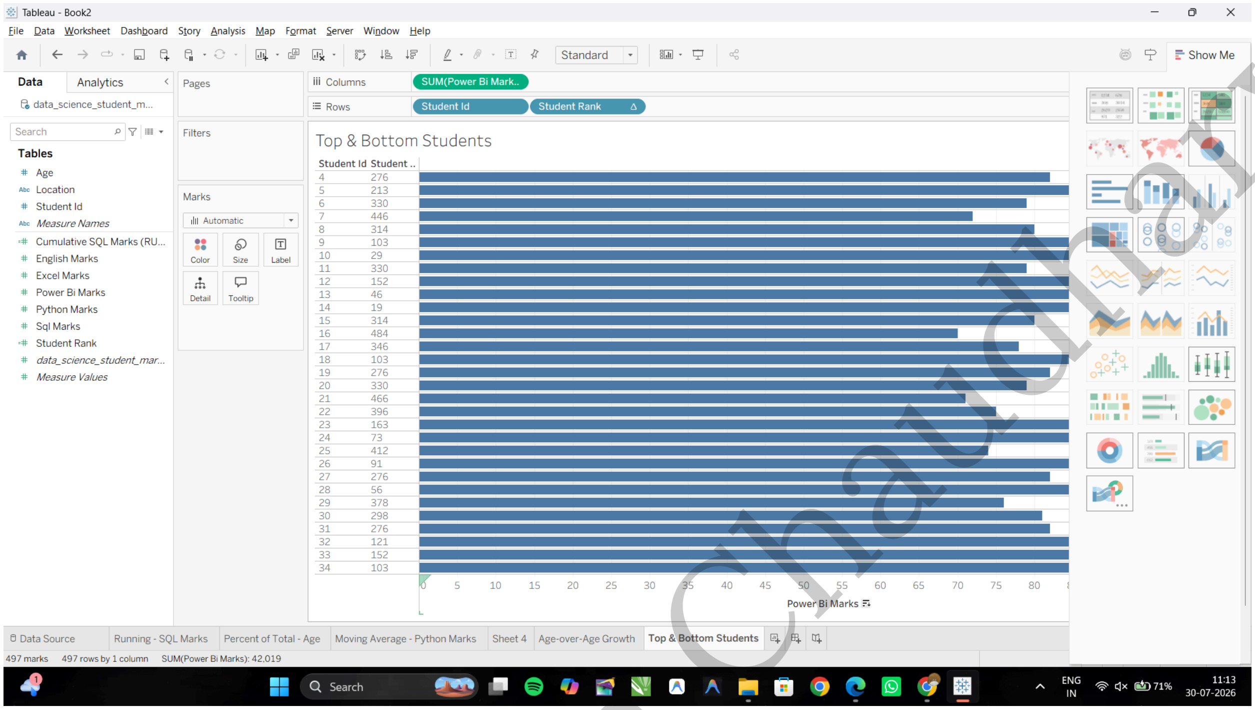
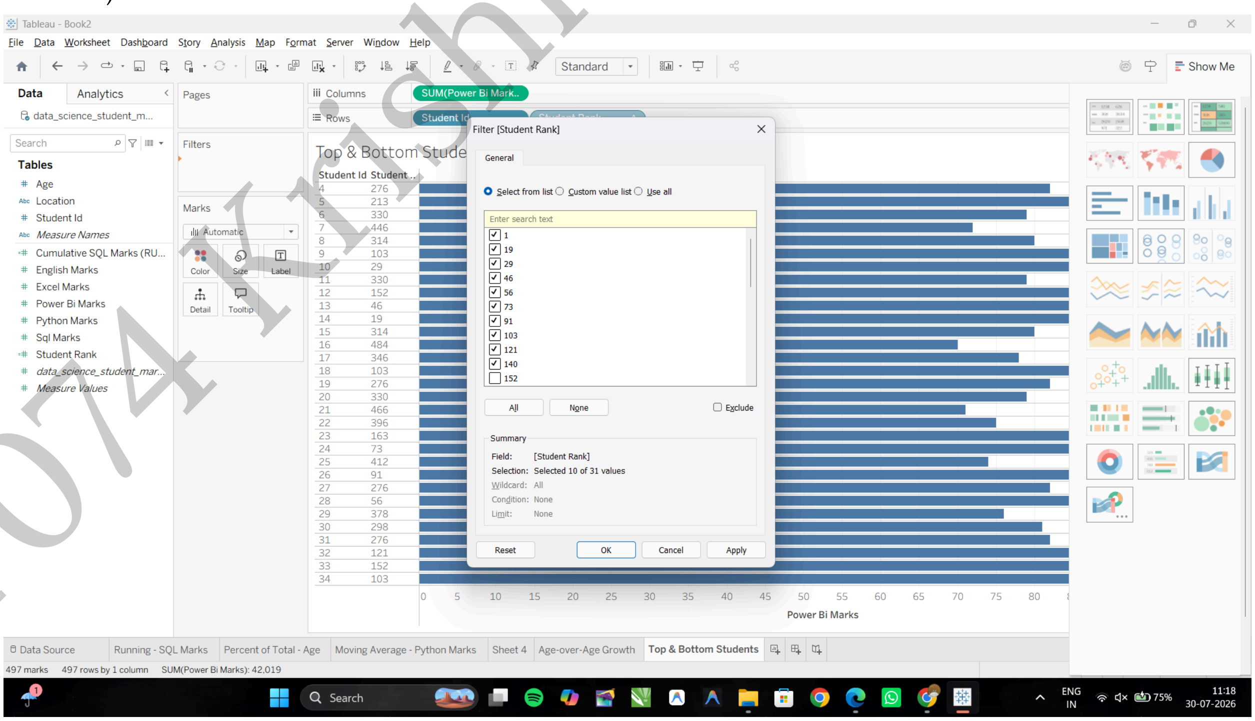
Task 3: Filter Dynamically to Display Top N Only

Hold Ctrl (or Cmd on Mac), click and drag the Student Rank pill from the Rows shelf, and drop it onto the **Filters** card.



In the filter dialog box, select a range of ranks (e.g., set the maximum to 10 to show only the Top 10 students).



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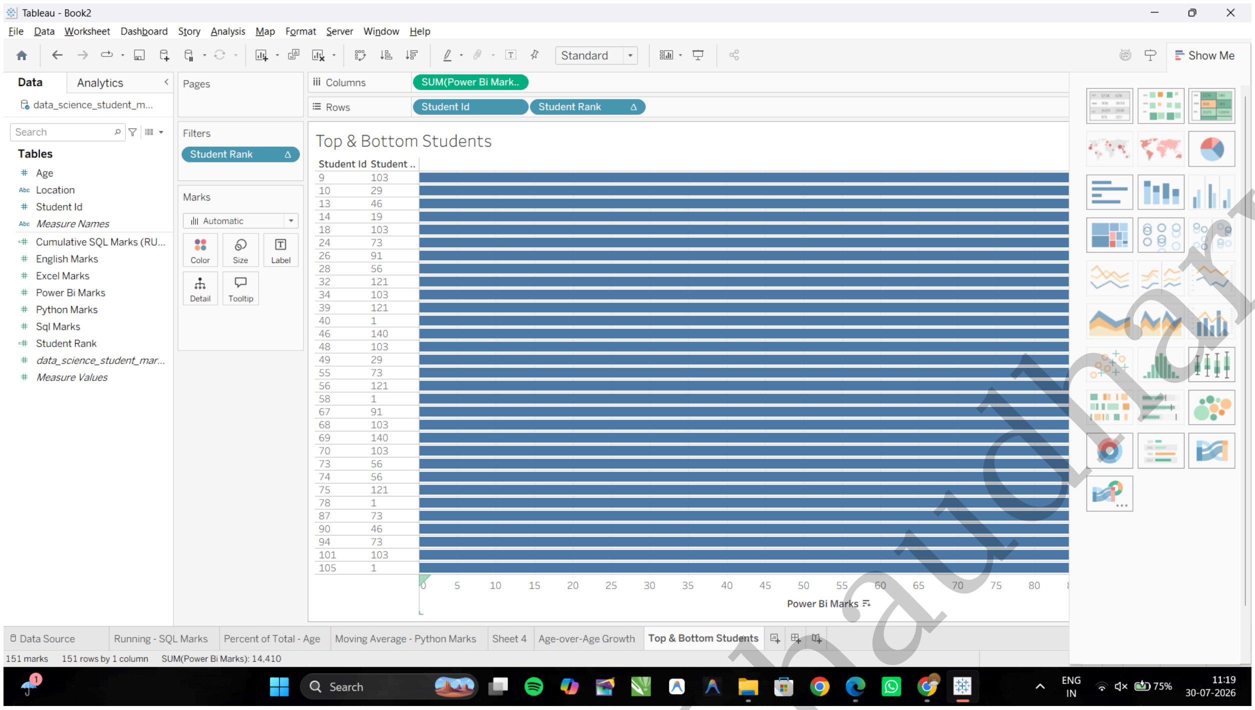
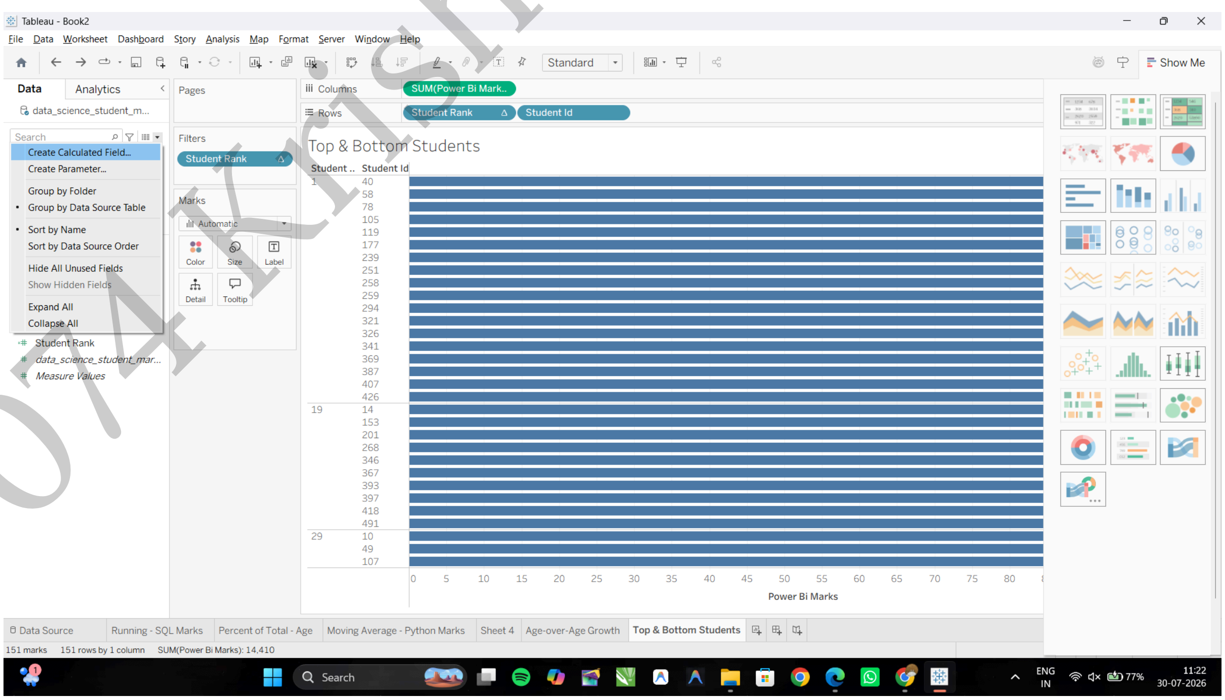


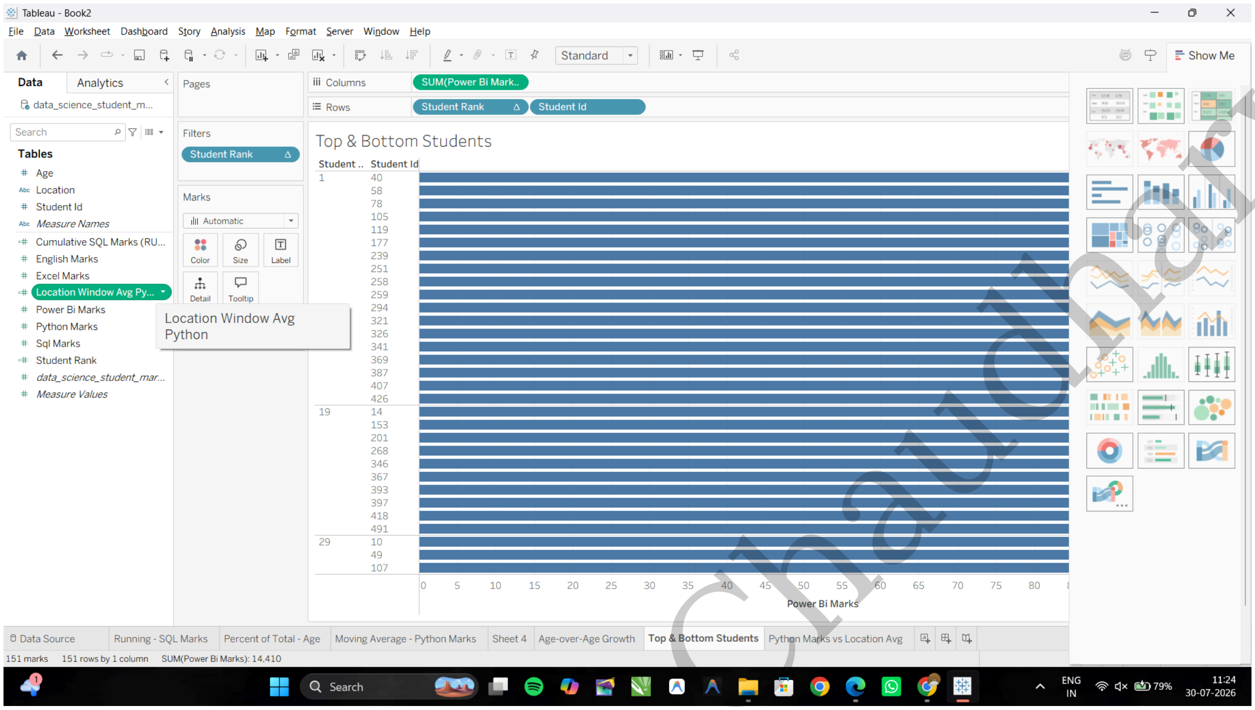
E: Implement window functions (WINDOW_SUM, WINDOW_AVG)

Task 1: Create a Window Average Calculation

Click the small drop-down arrow at the top right of the Data Pane (next to the Search bar) and select Create Calculated Field...

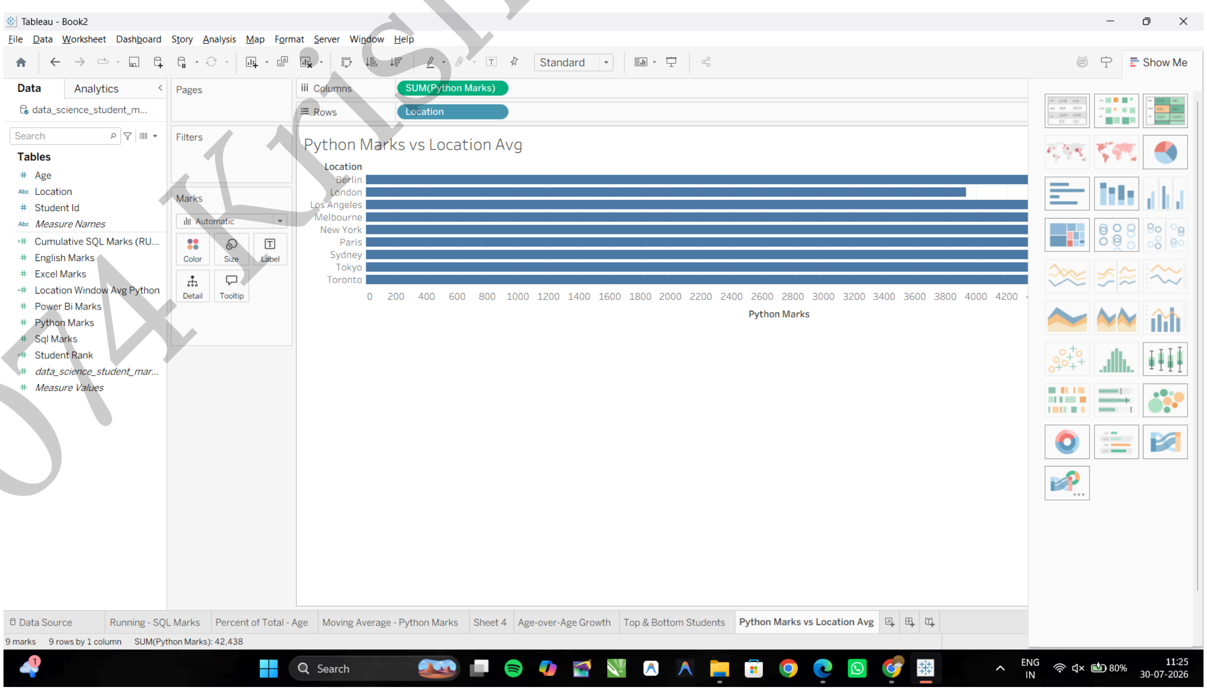


Name the field: Location Window Avg Python and input the following window function formula: WINDOW_AVG(AVG([Python Marks]))

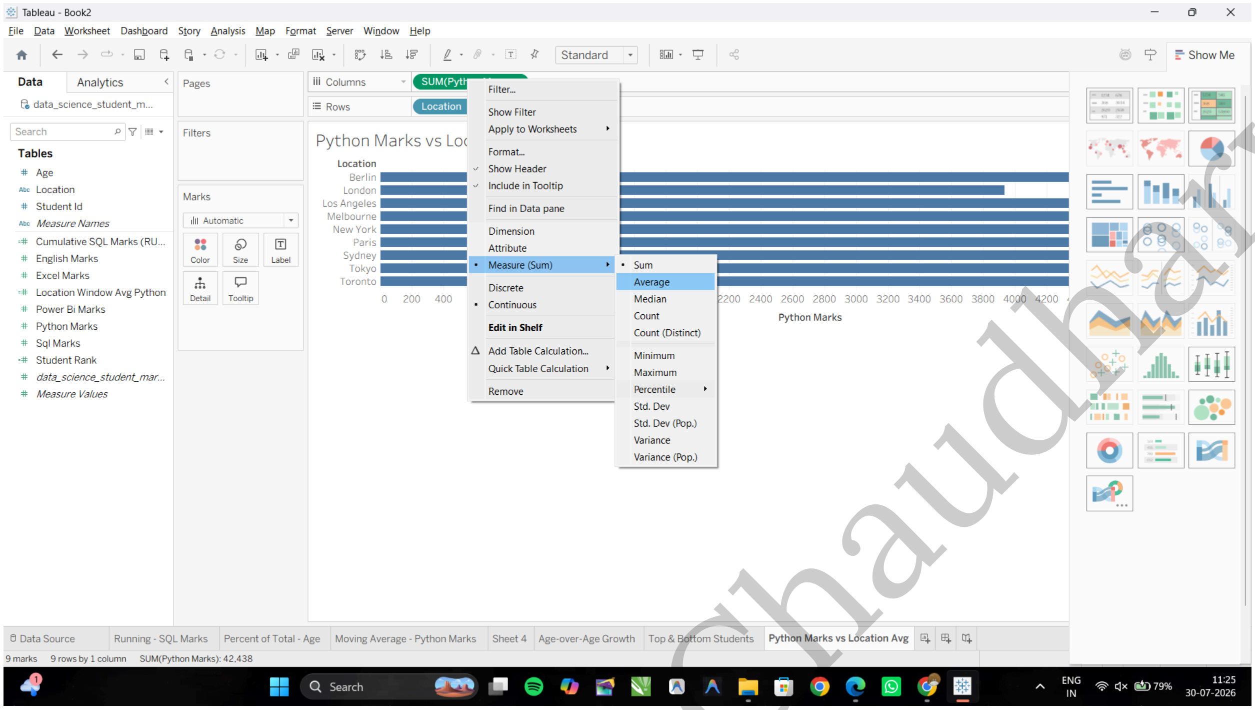


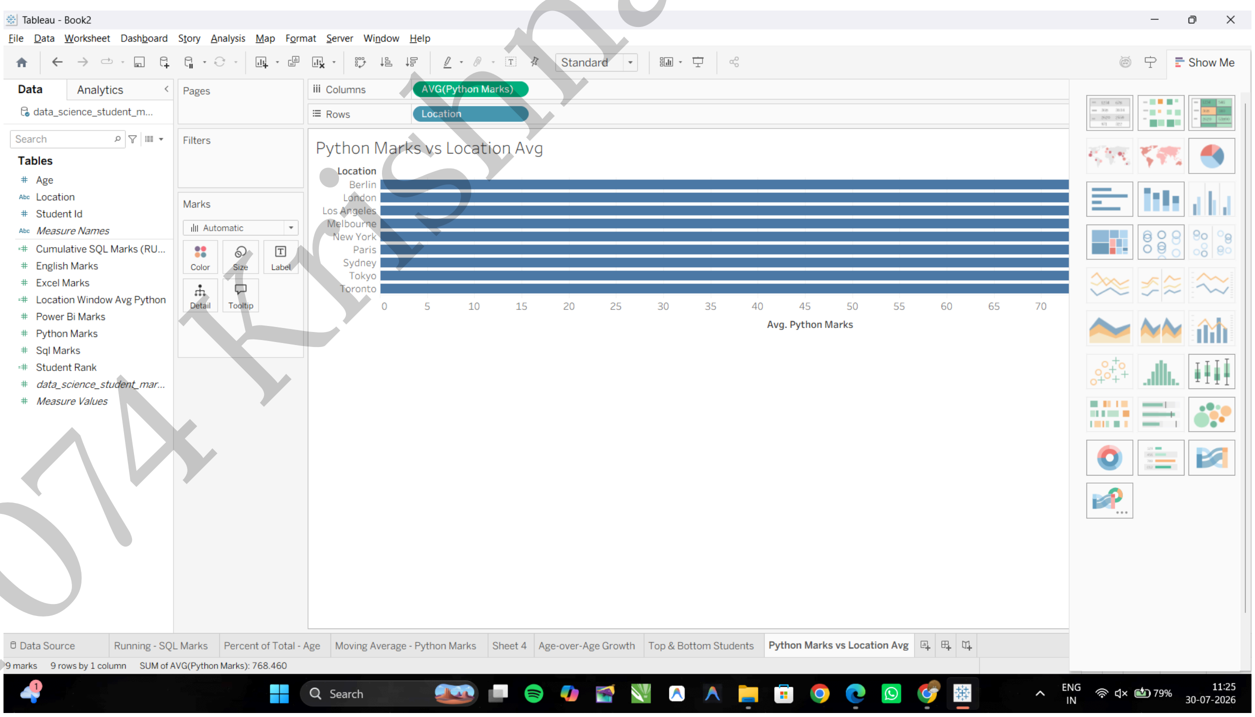
Task 2: Build the Window Comparison Visualization

Create a New Worksheet: Rename it Python Marks vs Location Avg, drag Location to the Rows shelf, and Python Marks to the Columns shelf.



Change its aggregation to Average (Right-click pill → Measure (Sum) → Average).





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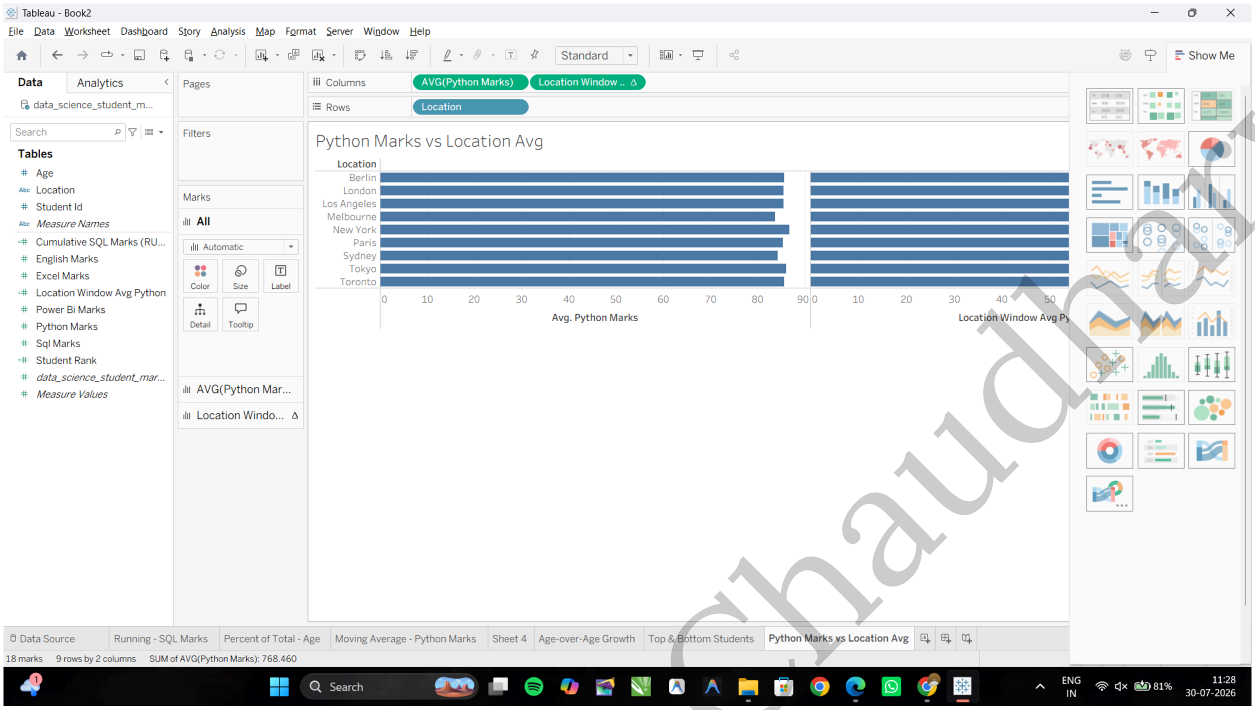
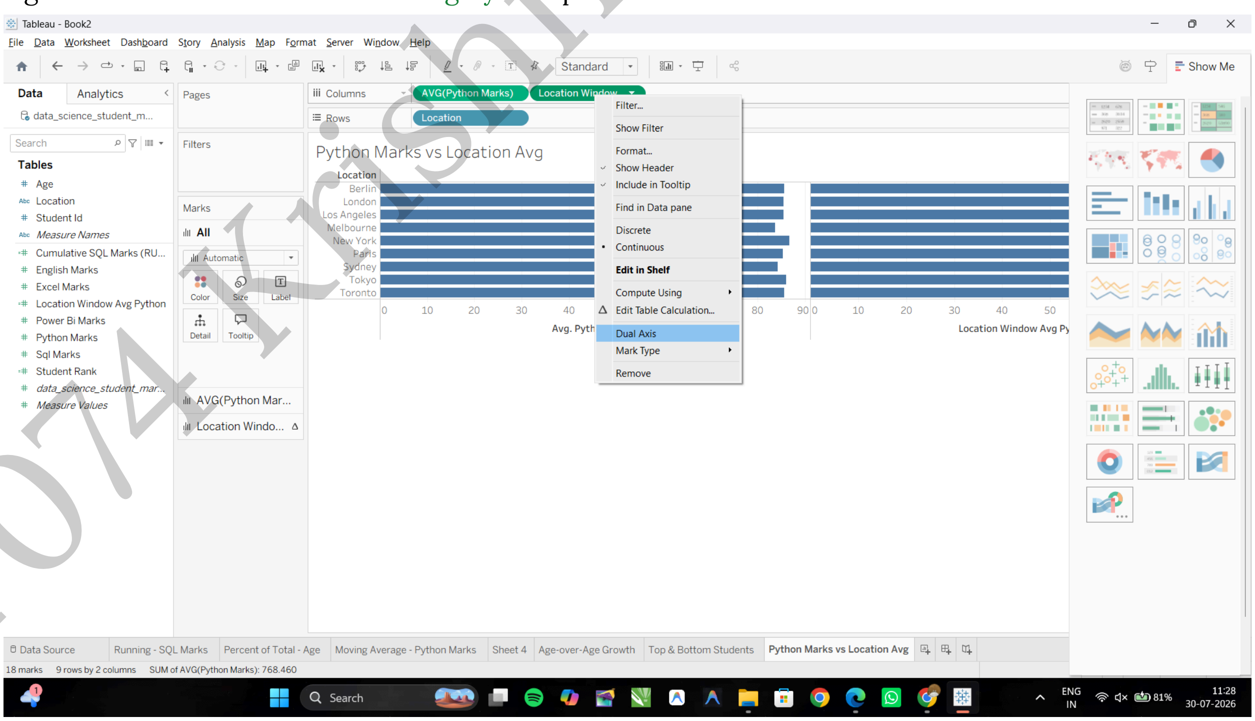
Drag newly created Location Window Avg Python calculated field onto the Columns shelf (placing it right beside AVG(Python Marks)).



Task 3: Combine Measures into a Dual-Axis Reference View

Right-click the Location Window Avg Python pill on the Columns shelf and select **Dual Axis**.



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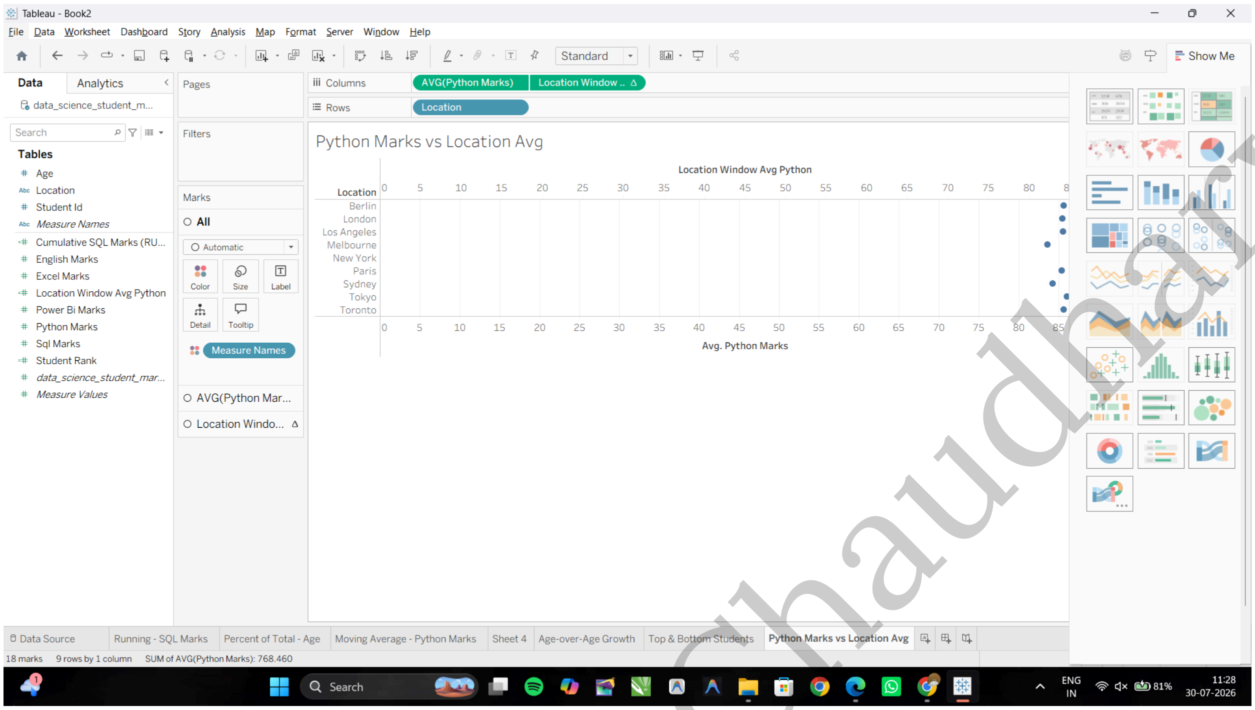
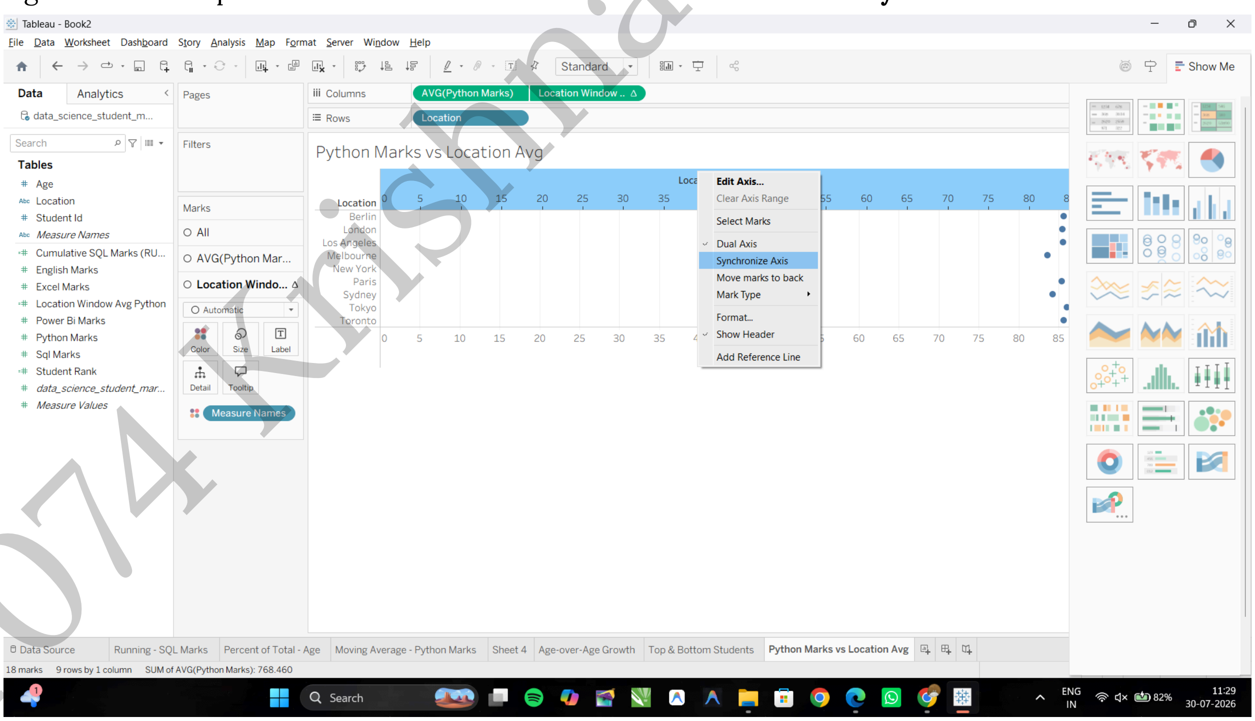


Right-click the top horizontal axis in the visualization view and select **Synchronize Axis**.



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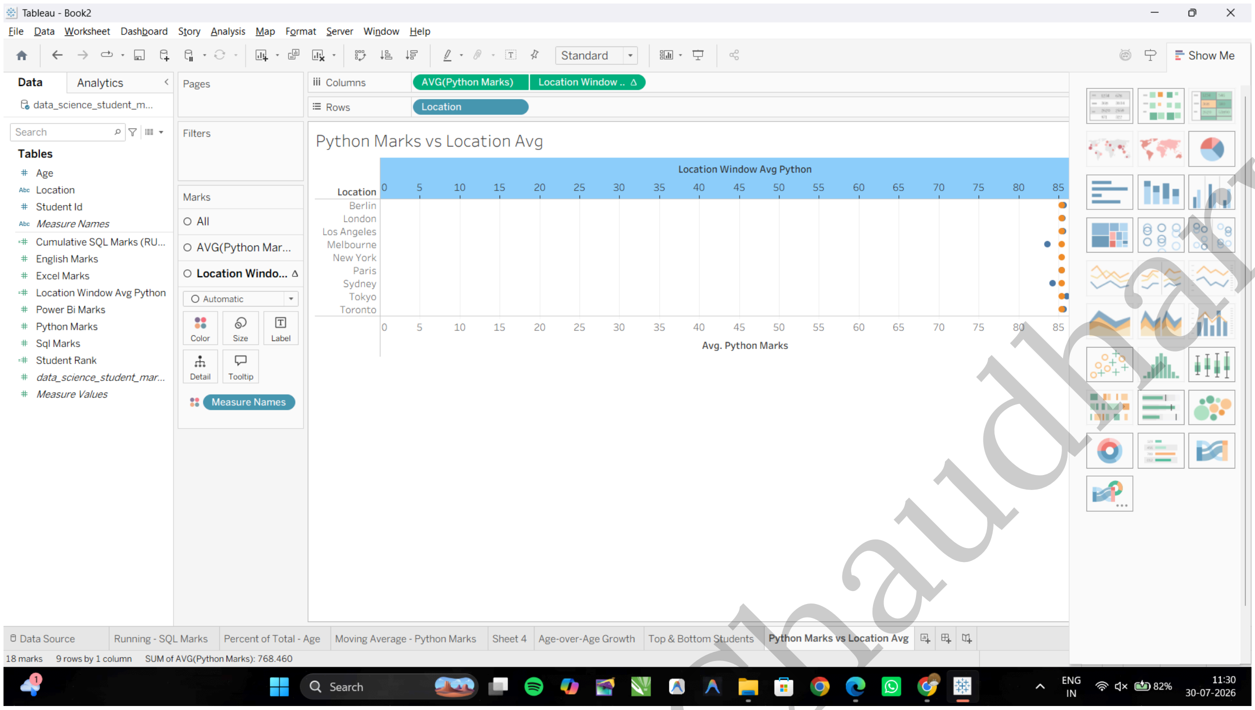
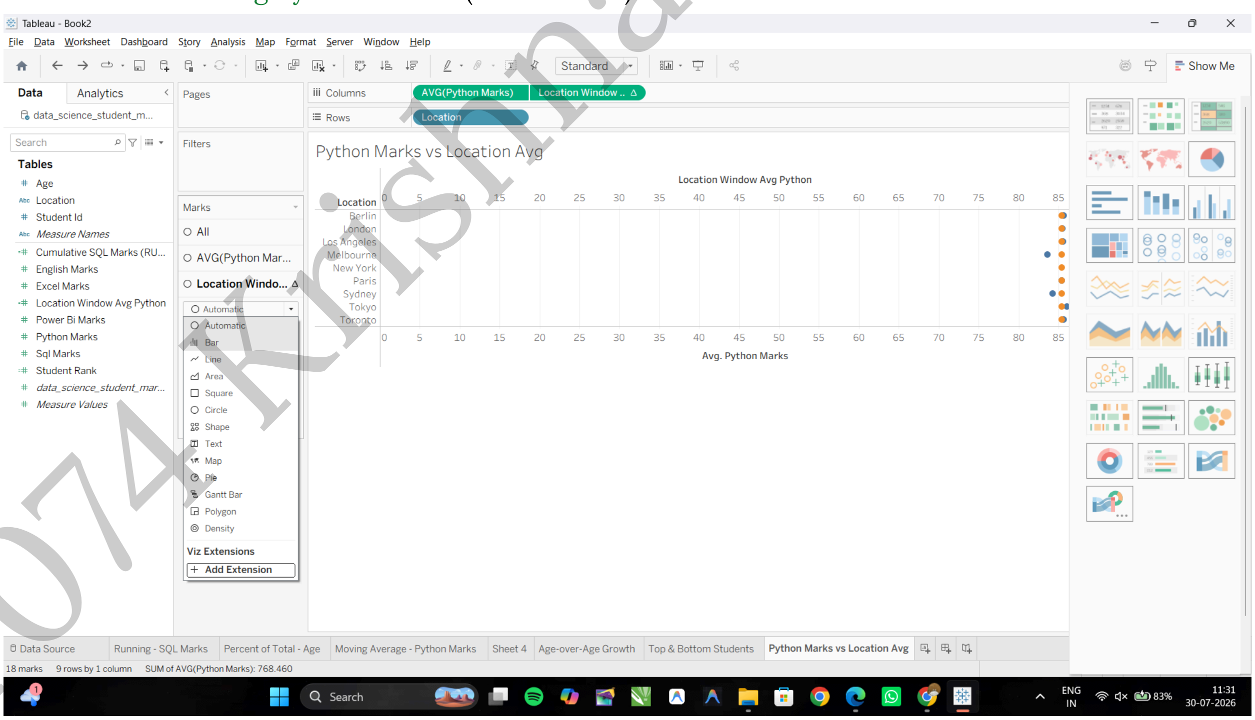


In the Marks card: Change the mark type for AVG(Python Marks) to Bar. Change the mark type for Location Window Avg Python to Line (or Gantt Bar).



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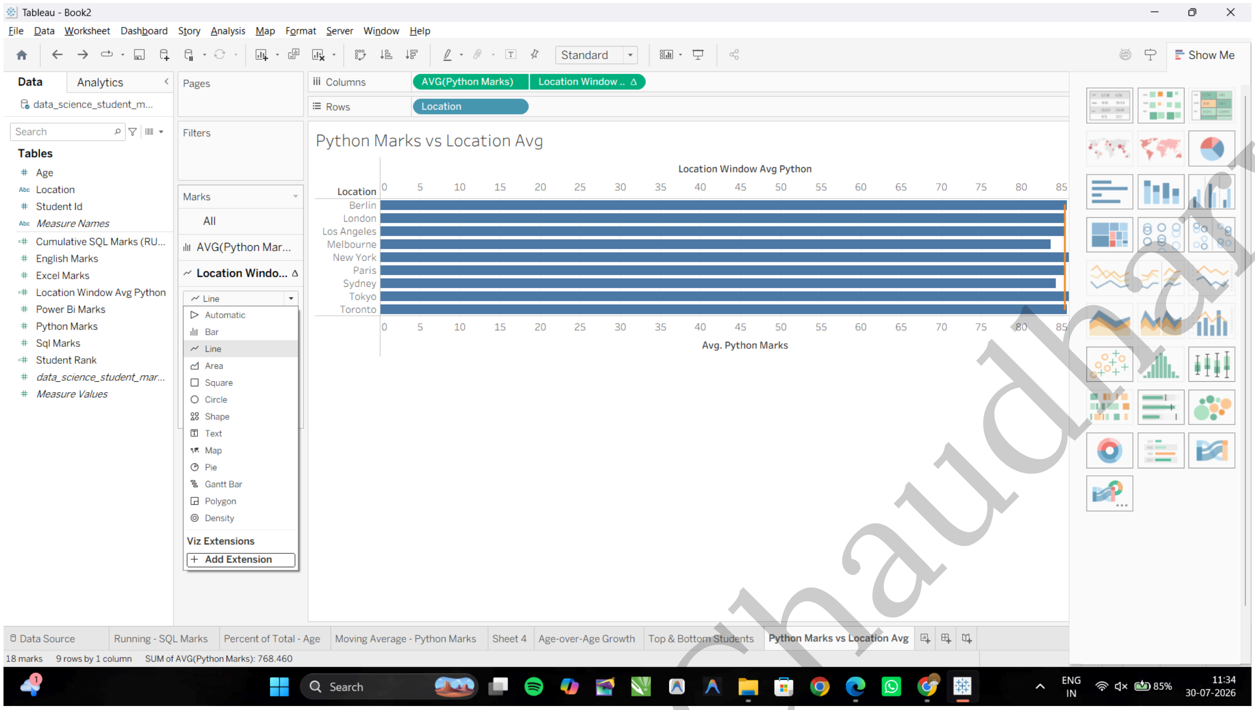

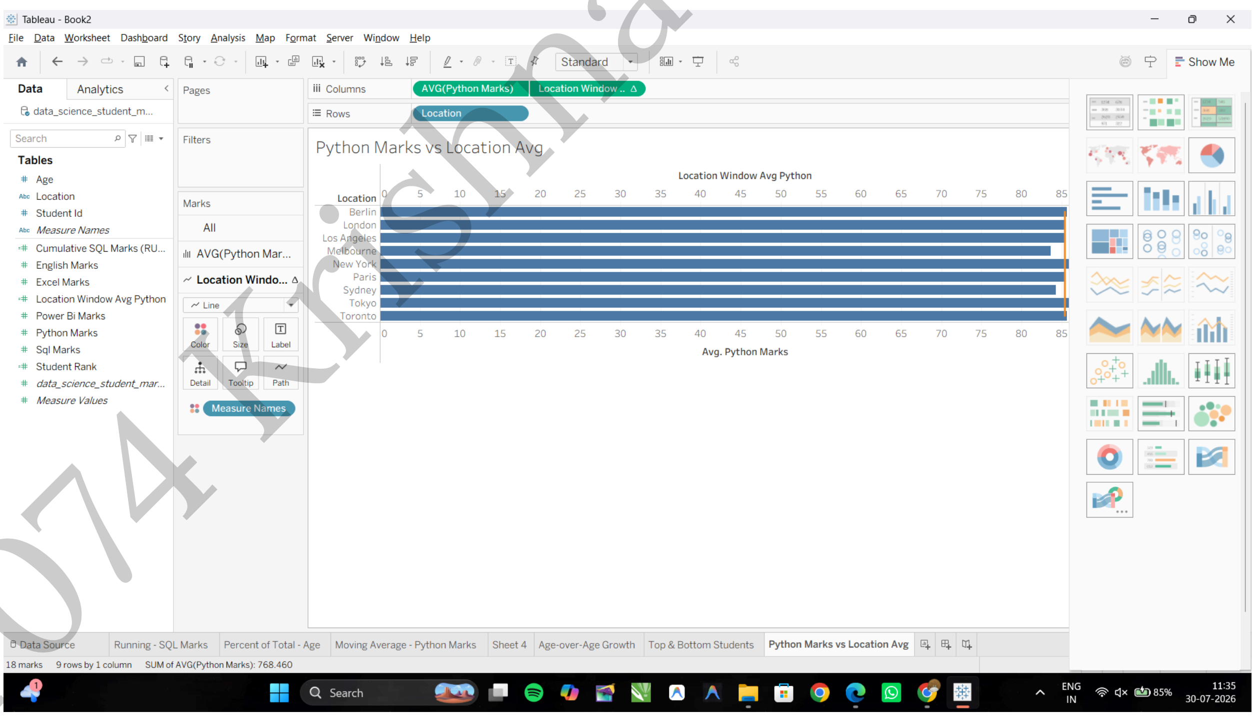

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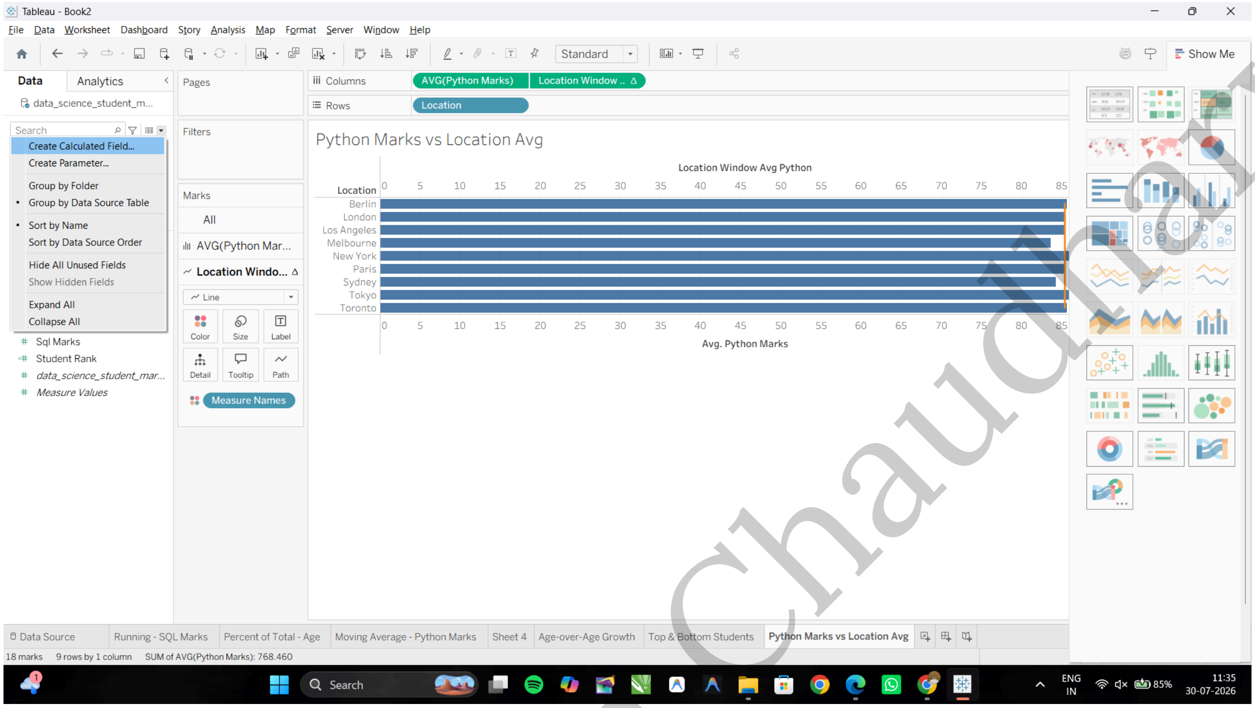
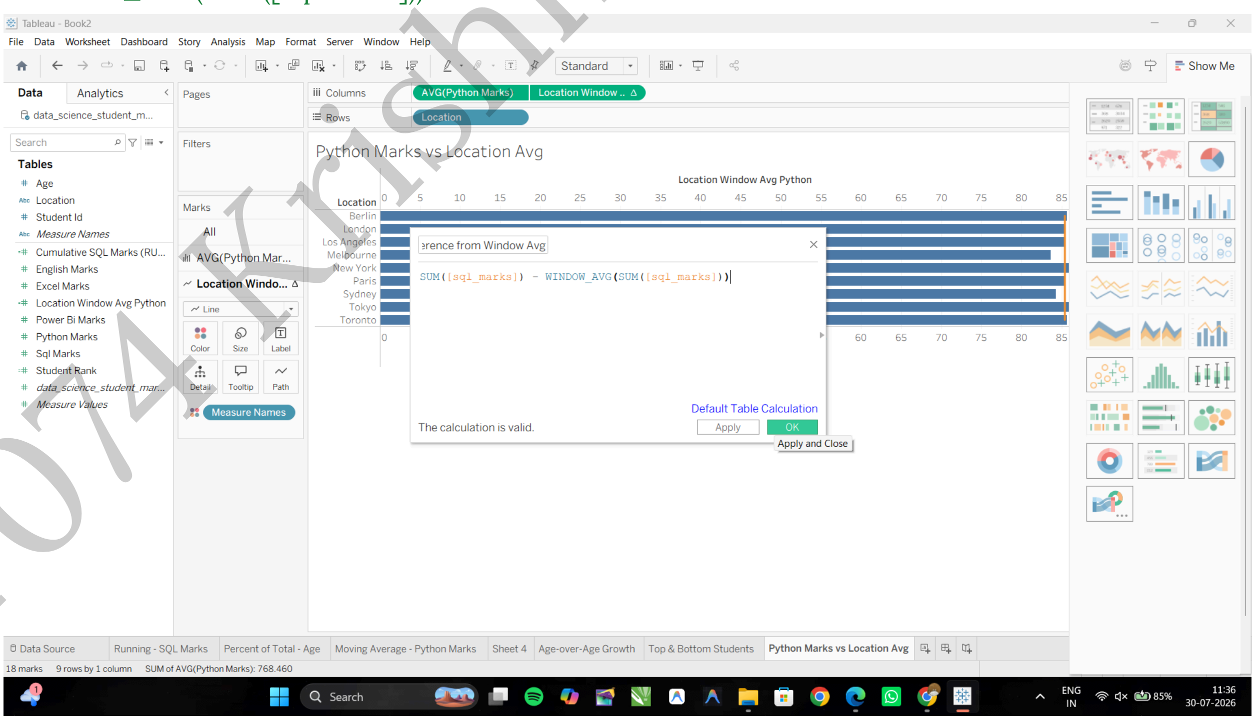
Task 4: Create a Window Difference Metric

Click the small drop-down arrow at the top right of the Data Pane and select Create Calculated Field...



Name the field: Difference from Window Sum. Input the following expression to compare each mark count against the total sum of all marks in the visible window: SUM([Sql Marks]) - WINDOW_AVG(SUM([Sql Marks]))



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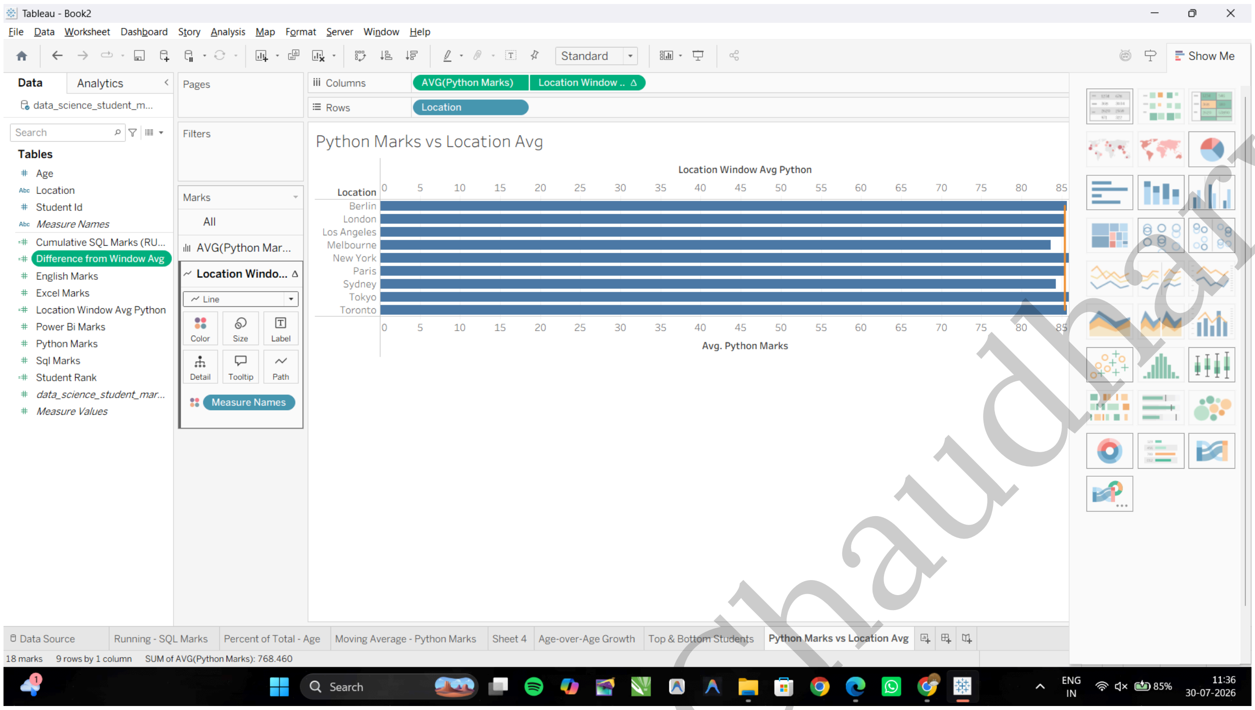
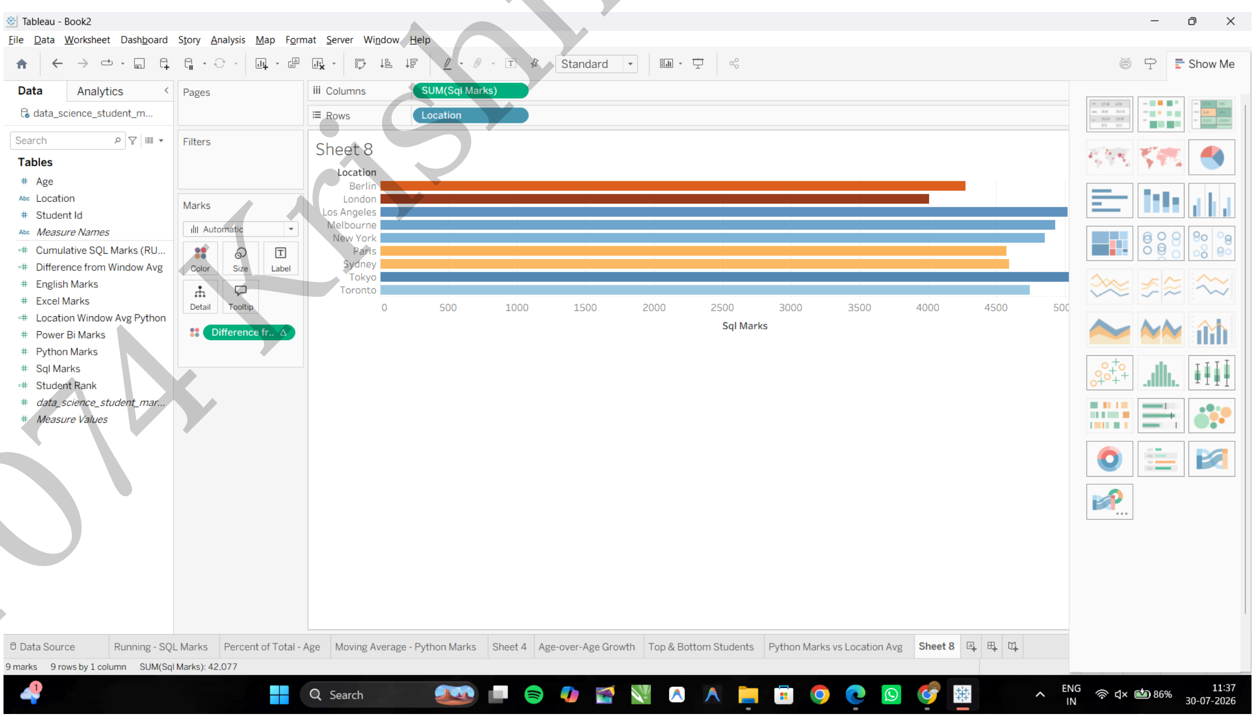


Apply it to the View: Create a New Worksheet: Drag Location to the Rows shelf, Sql Marks to the Columns shelf, then drag the Difference from Window Sum calculated field to Color on the Marks card.



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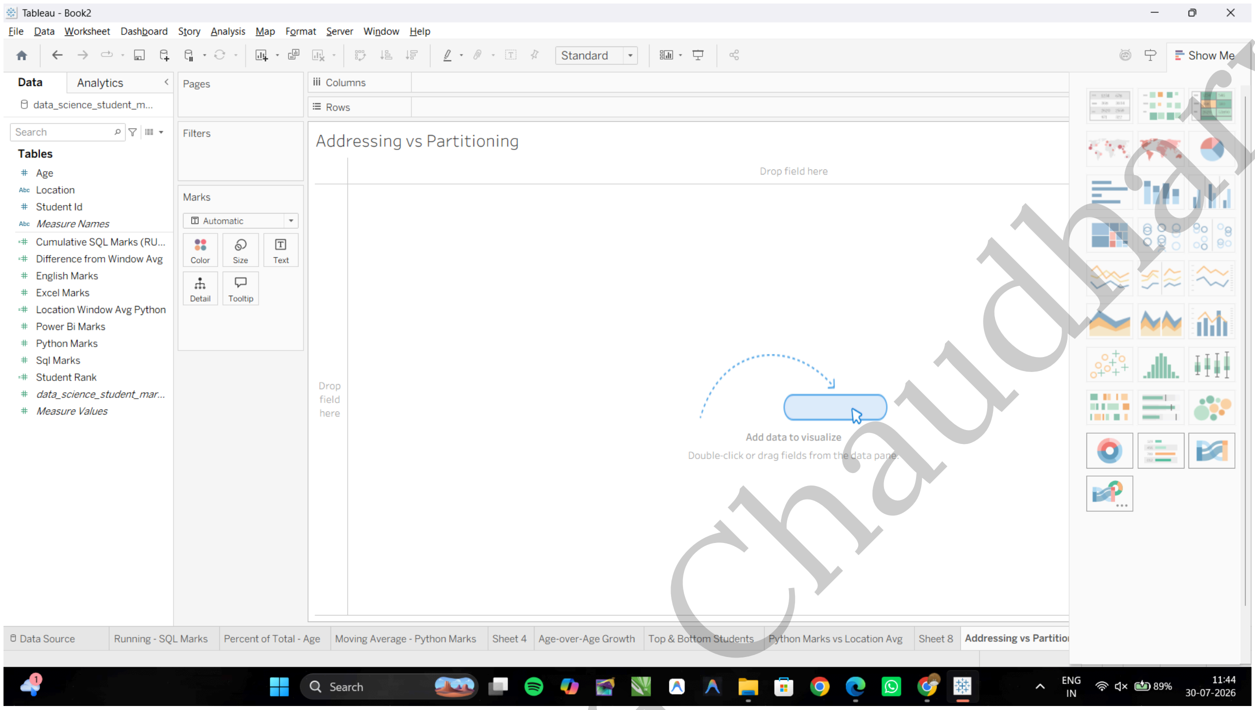
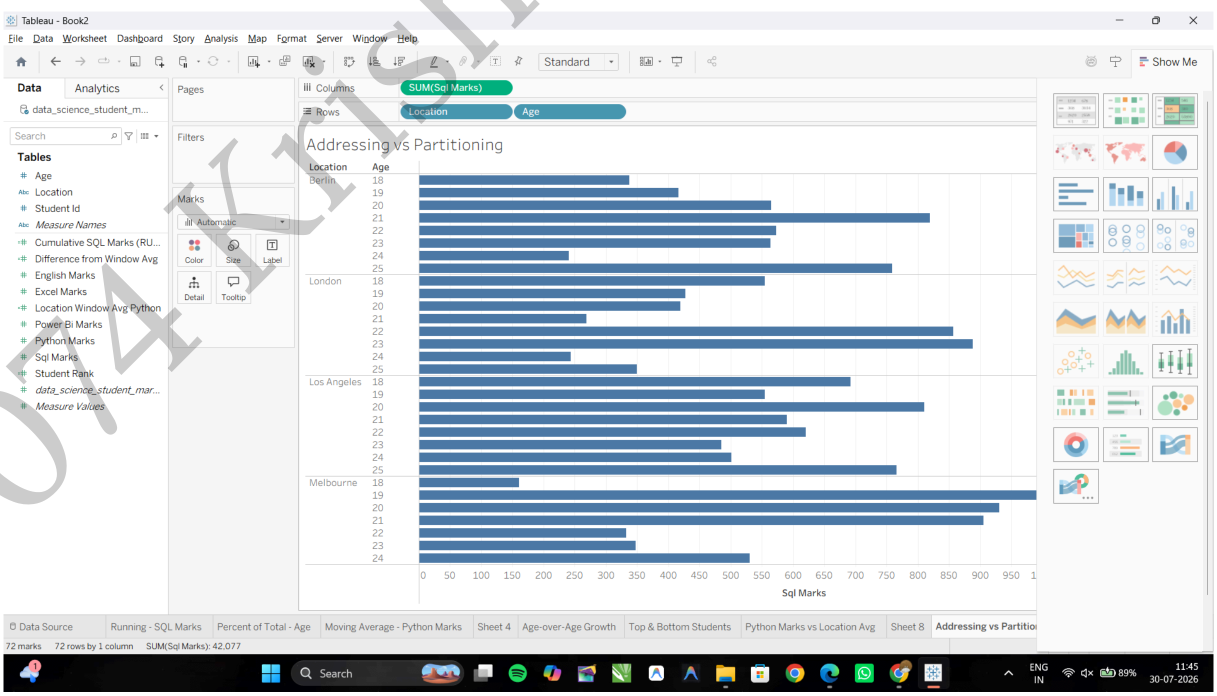
F: Customize addressing and partitioning

Task 1: Build a Nested View for Location and Age

Open a new worksheet and rename it: Addressing vs Partitioning



Create a Nested View: Drag **Location** and **Age** to the **Rows** shelf (Age next to Location), then drag **Sql Marks** to the **Columns** shelf as **SUM(Sql Marks)**.



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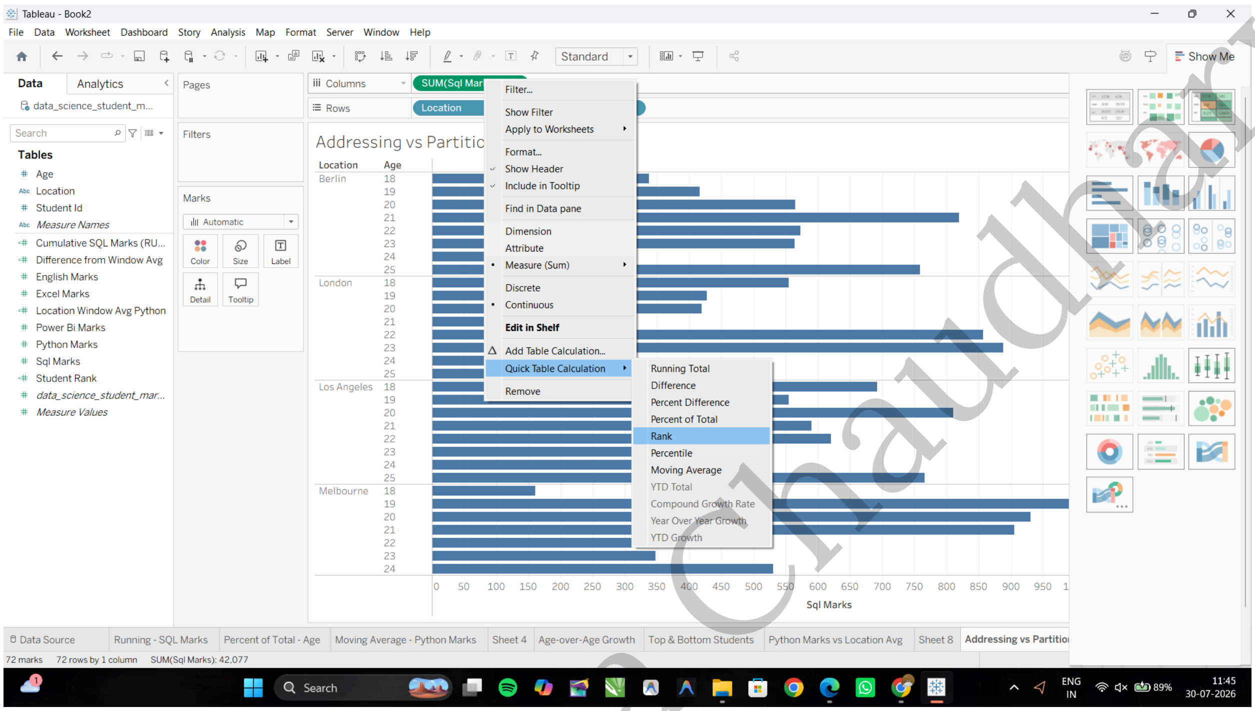
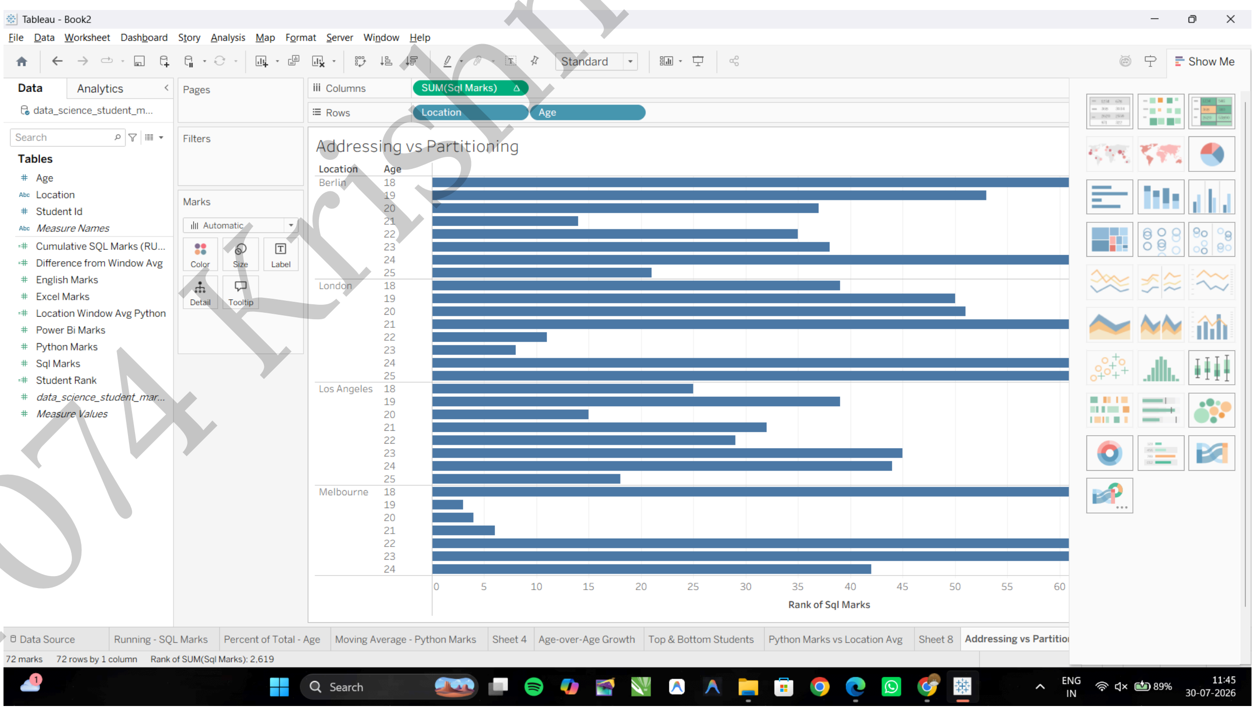
Task 2: Apply Partitioning via "Compute Using" Options

Right-click the SUM(Sql Marks) pill on the Columns shelf, hover over Quick Table Calculation, and select Rank.





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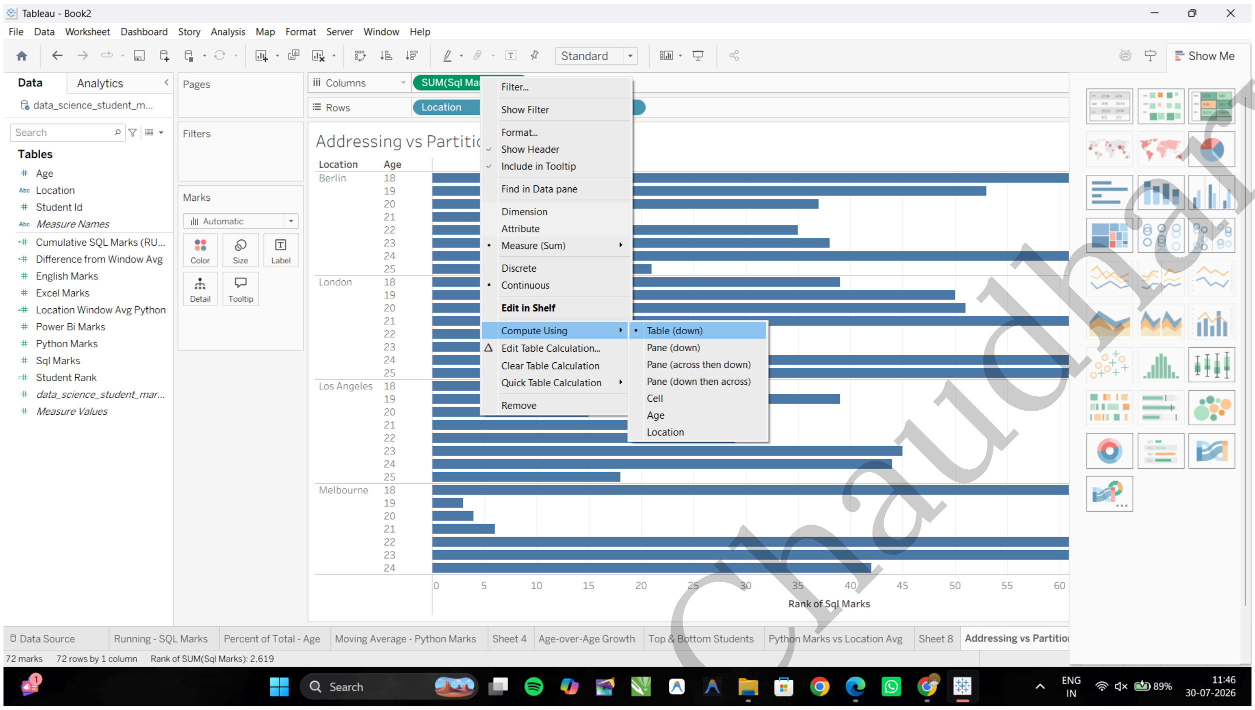
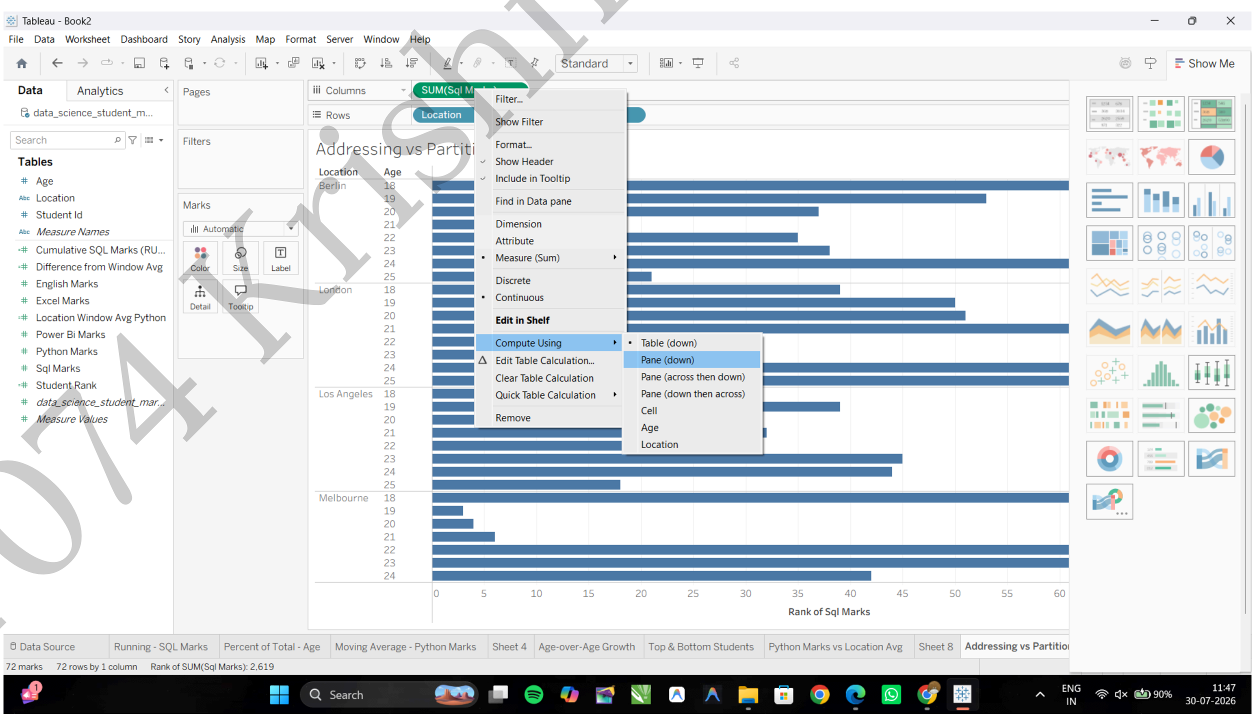
Compute Using: Right-click SUM(Sql Marks) → Compute Using → Select Table (Down) for the entire table, or Pane (Down) to restart the ranking for each Location.





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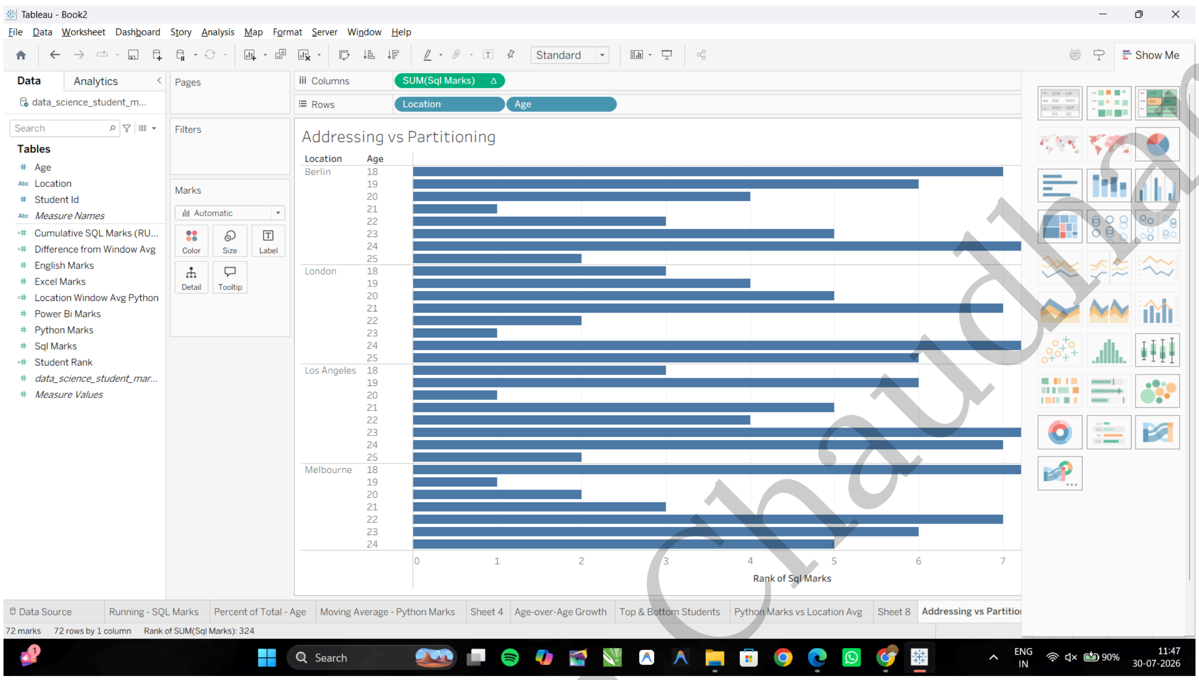
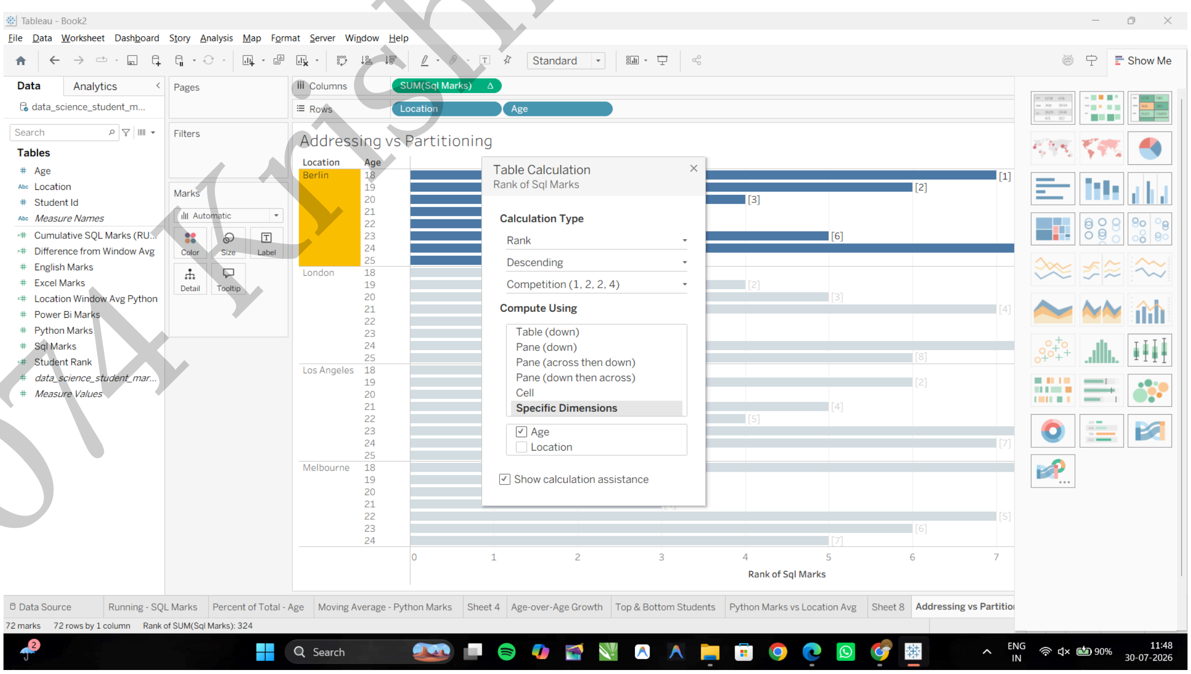
Task 3: Manually Define Addressing with Specific Dimensions

Right-click the SUM(Sql Marks) pill (with the Δ icon) on the Columns shelf and select Edit Table Calculation...



Specific Dimensions: Right-click **SUM(Sql Marks)** → **Compute Using** → **Specific Dimensions**. Uncheck **Location**, keep **Age** checked, then close the **Table Calculation** dialog box.



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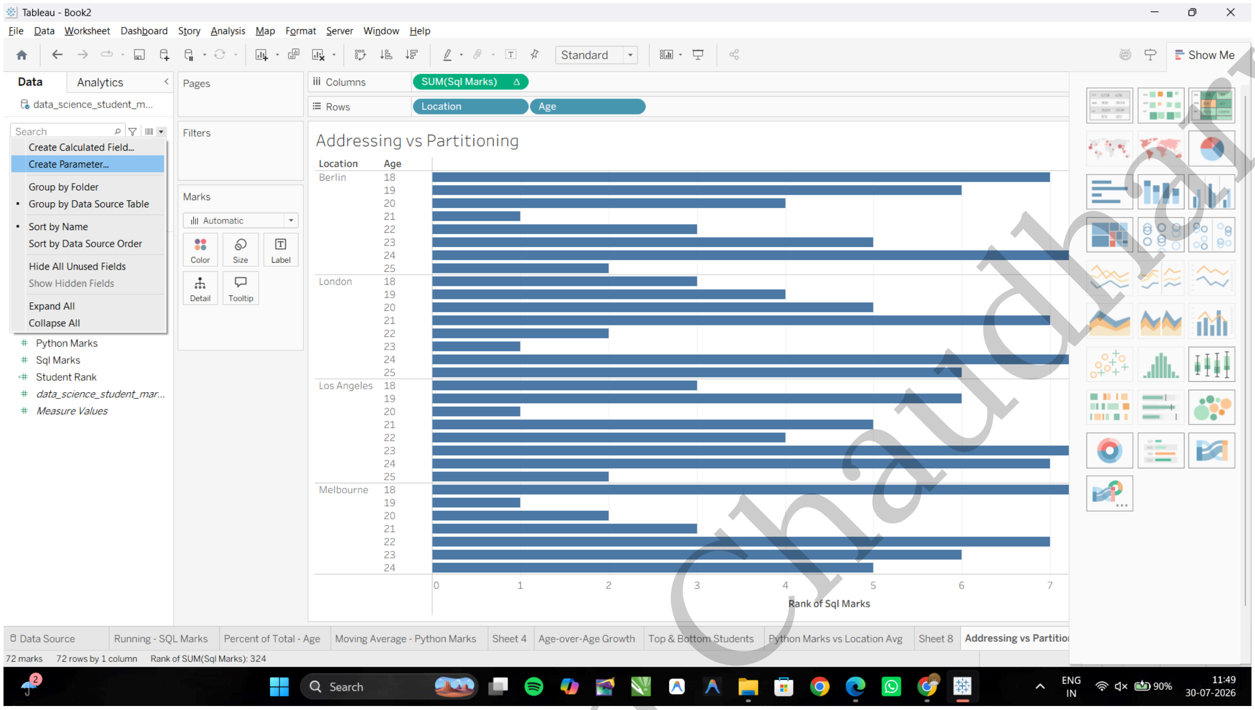
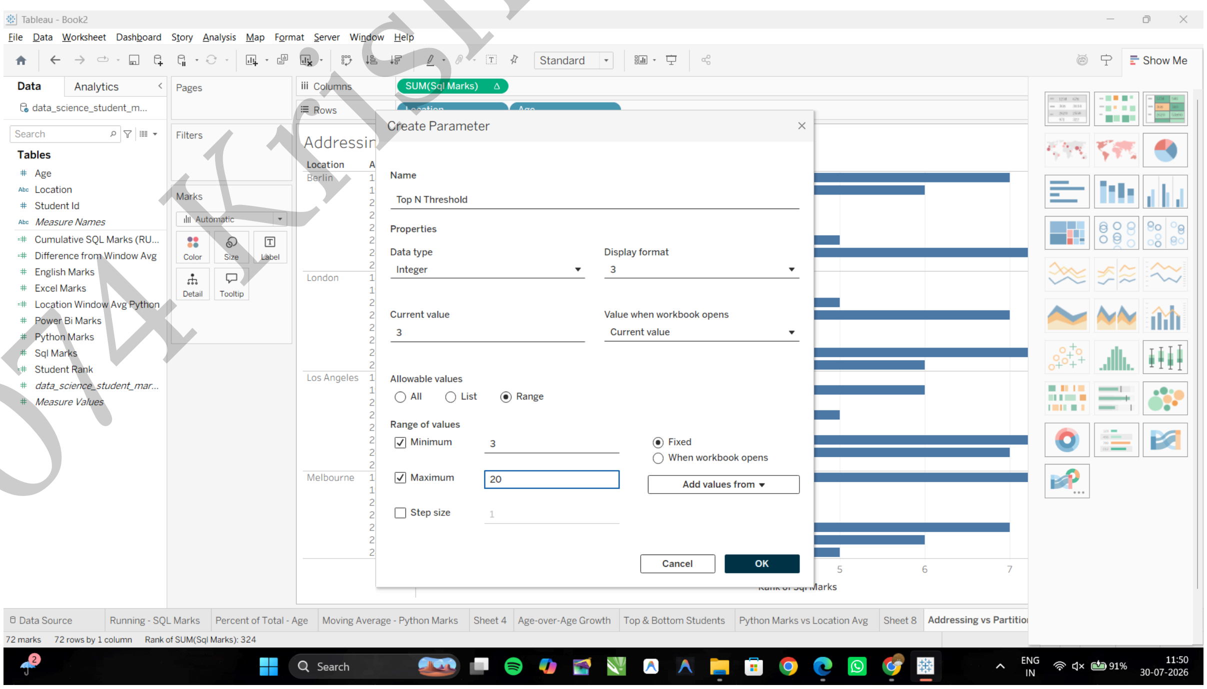
G: Create dynamic titles and labels using table calculations

Task 1: Create a Dynamic Label Parameter

Right-click anywhere in the Data Pane and select Create Parameter...



Create a Parameter: Name it Top N Threshold, set Data Type to Integer, Current Value to 5, choose Range (Min: 3, Max: 20, Step: 1), then click OK.



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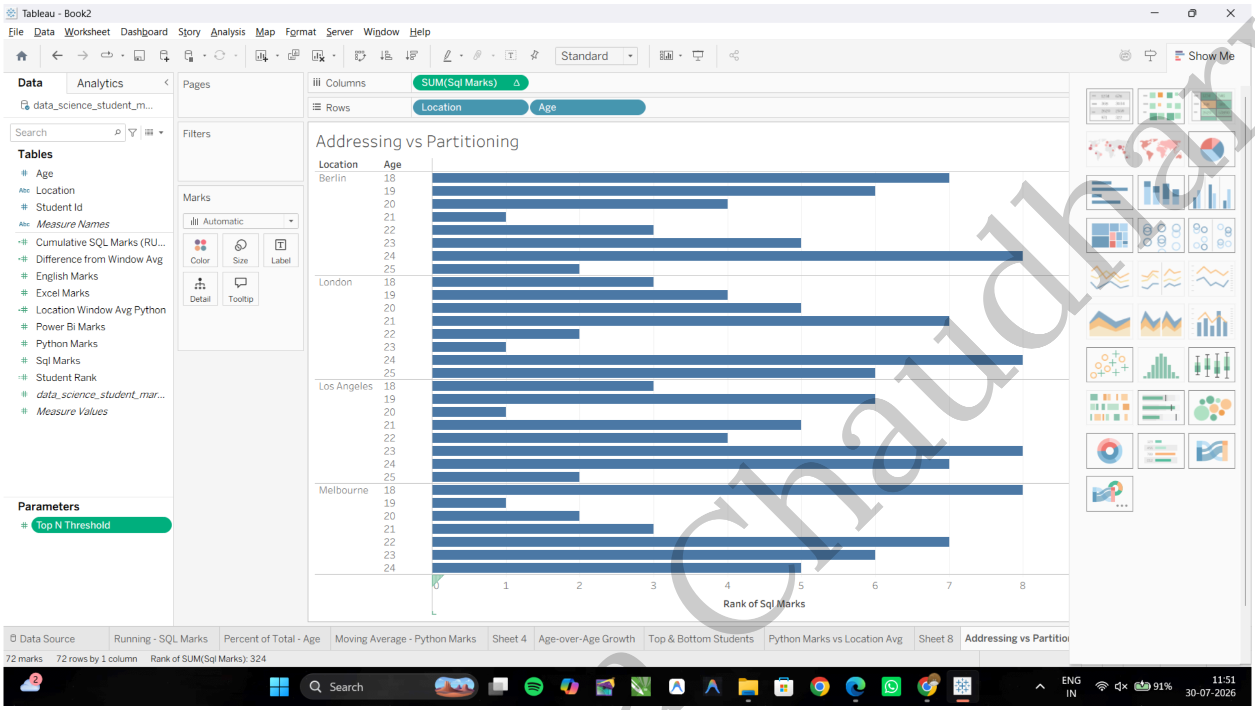
Task 2: Create a Dynamic Header Calculation

Create a new Calculated Field named Dynamic Title: "Top " + STR([Top N Threshold]) + " Students by Total SQL Marks"



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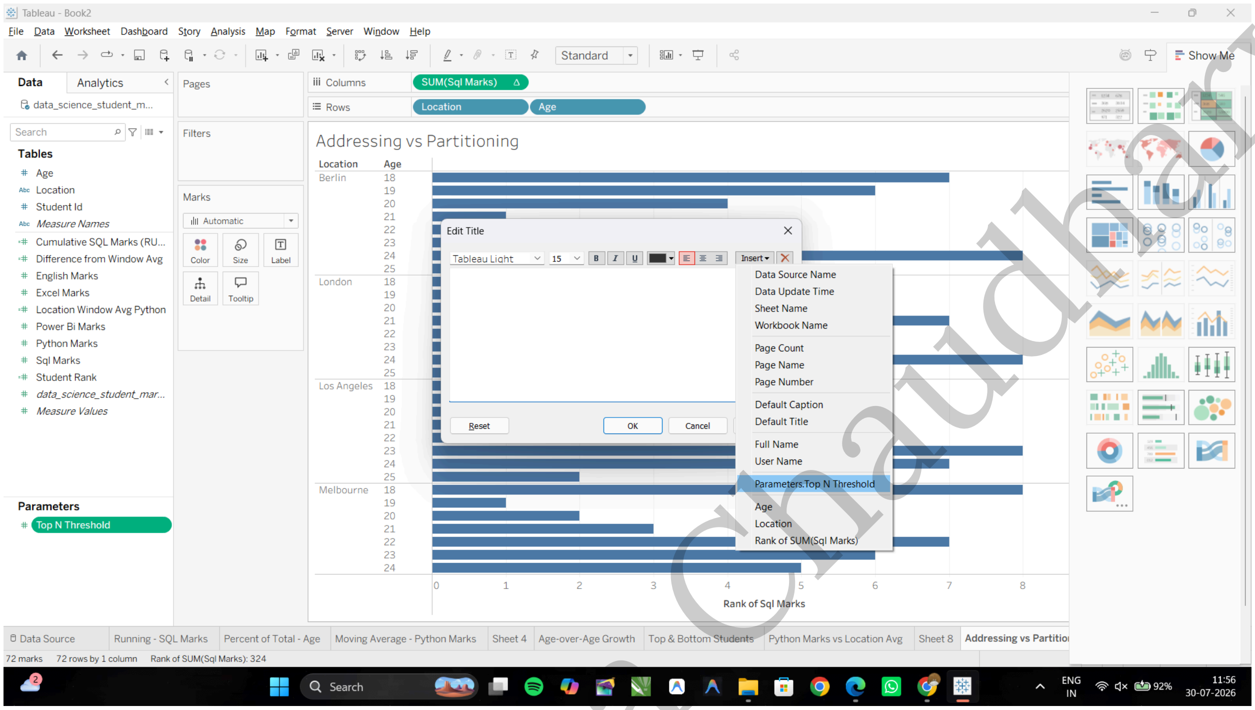
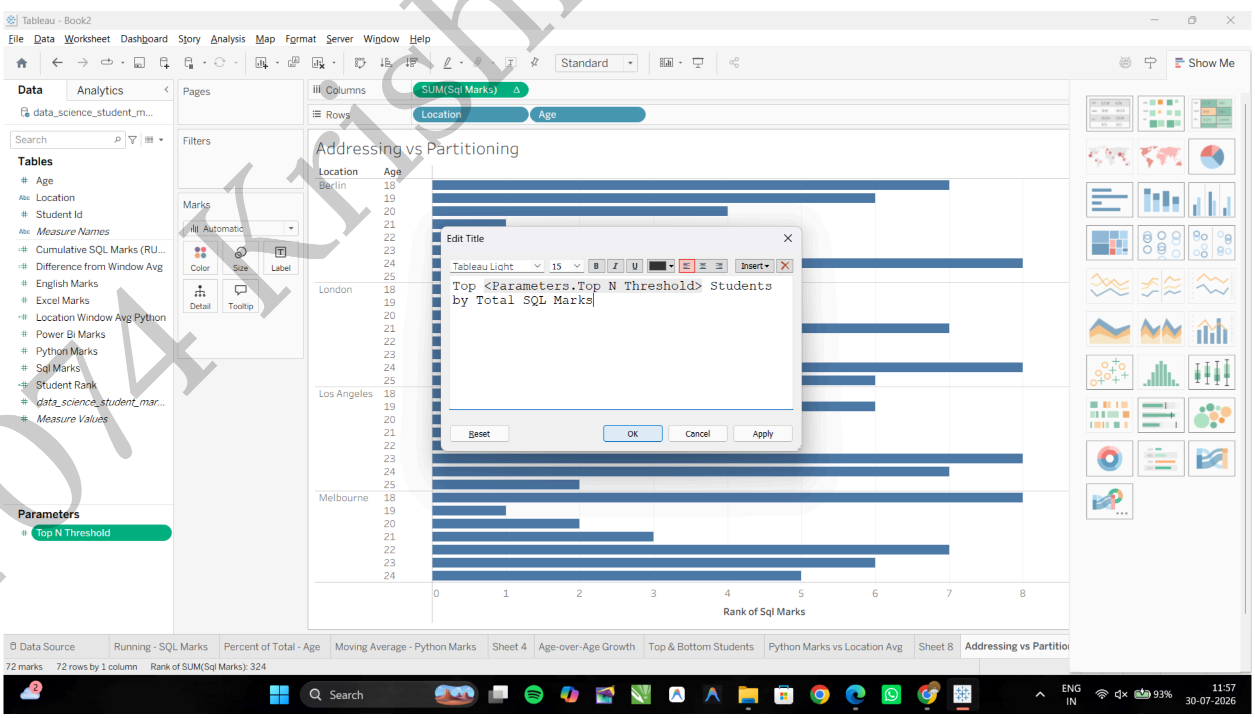
Task 3: Embed Calculated Fields and Parameters into the Sheet Title

Edit the Sheet Title: Double-click the title, clear the text, click **Insert** → **Parameters** → **Top N Threshold**, set the title to **Top <Top N Threshold> Students by Total SQL Marks**, then click **OK**.





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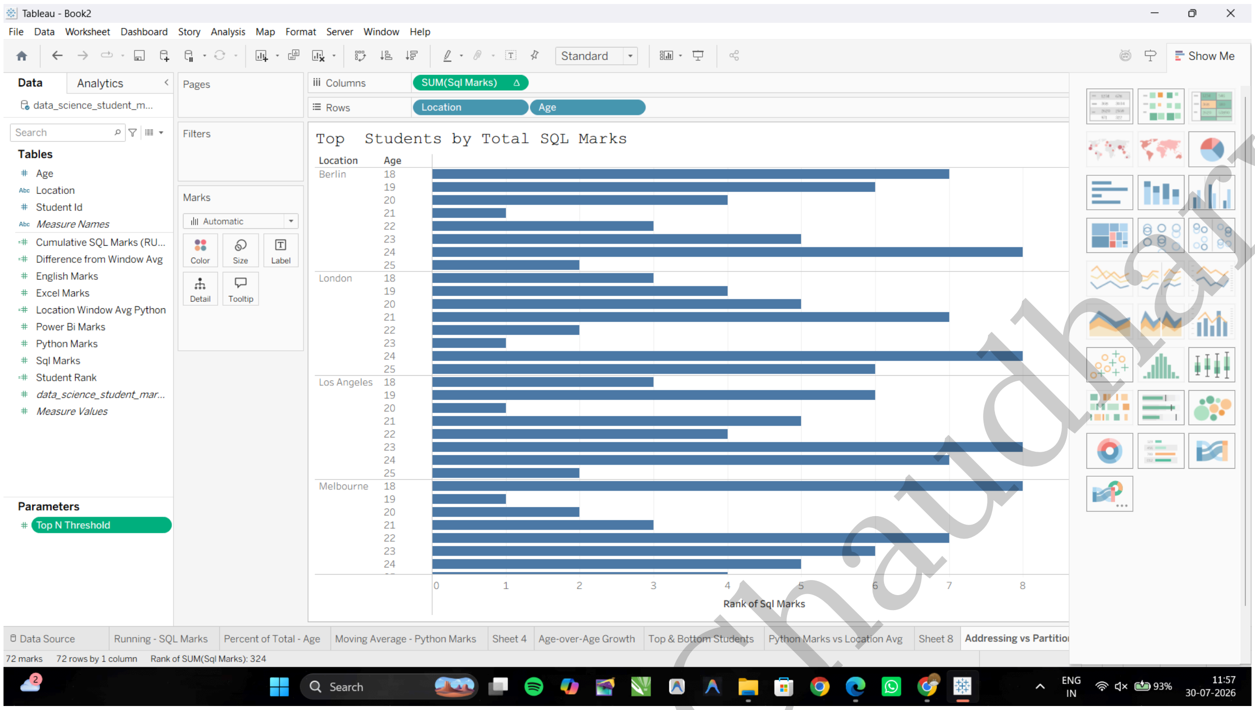
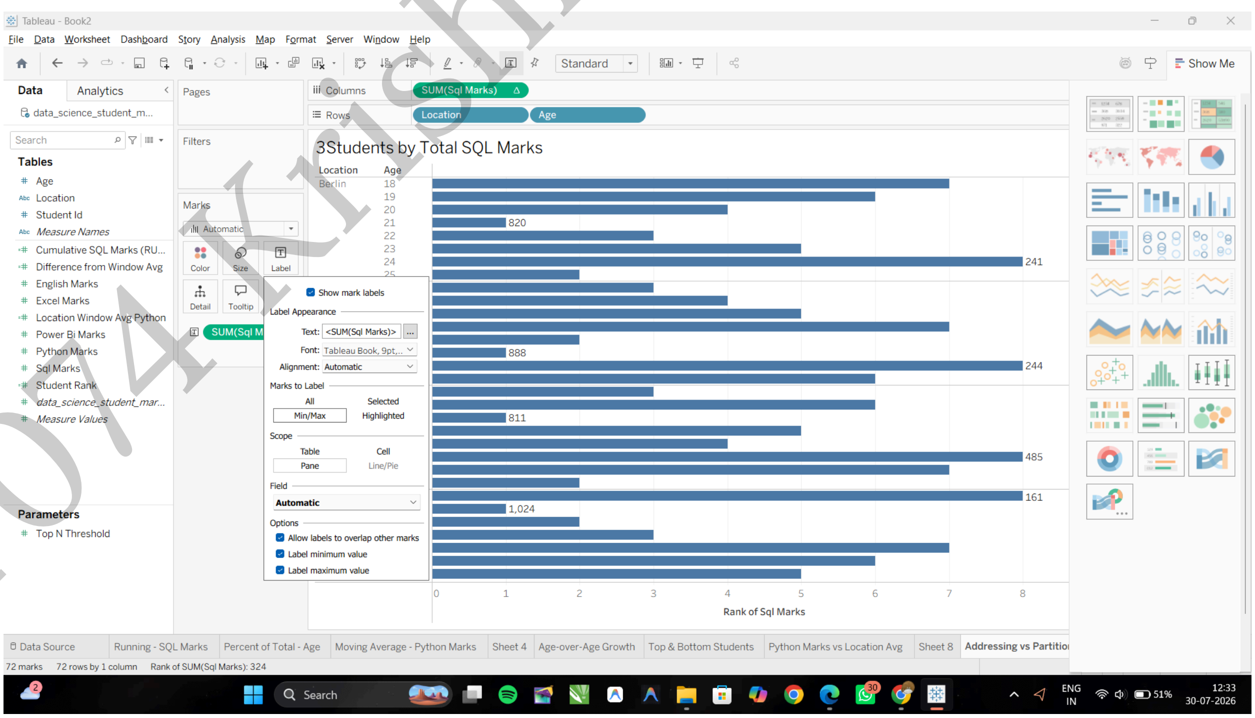


Task 4: Add Dynamic Mark Labels (Highlighting Extremes)

Add Labels: Drag SUM(Sql Marks) to Label. Open Label options, set Marks to Label to Min/Max, choose SUM(Sql Marks) as the field, and enable Allow labels to overlap other marks.



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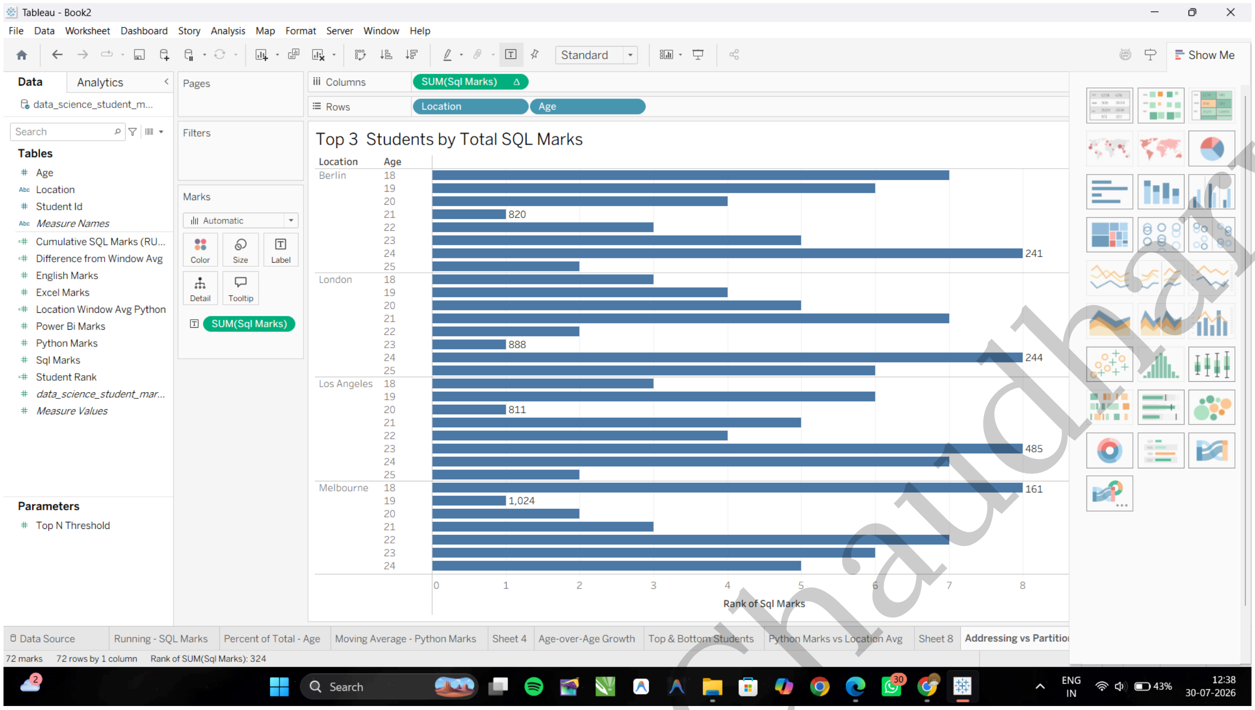

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